VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



CAPSTONE PROJECT

Project:

Analyzing the Role of Image Captioning in Enhancing Text-to-Image Retrieval Performance

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Major: Computer Science

THESIS COMMITTEE: Committee 9CC - Computer Science
SUPERVISOR: PhD. Le Thanh Sach

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STUDENT 1: Vu Minh Quan (2212828)
STUDENT 2: Dang Huu Nhat Hoang (2211069)
STUDENT 3: Nguyen Anh Lam (2211795)

Ho Chi Minh City, May 2026

DECLARATION

We hereby declare that this Capstone Project is the result of our collective group study, research, and implementation, conducted under the direct guidance of the supervisor. The content of this thesis has been developed based on the knowledge acquired during our academic training, a thorough review of relevant scientific literature, and experimental results carried out by our group.

All data, figures, tables, and experimental results presented in this report are truthful, objective, and accurately reflect the research process. Any content referenced or adapted from external sources, including books, scientific articles, and online materials, has been properly cited and clearly acknowledged in the list of references in accordance with academic standards.

We further declare that this project has not been submitted, in whole or in part, for the fulfillment of any degree, course requirement, or academic evaluation at any other institution. We fully understand that any violation of academic integrity or plagiarism, if discovered, will result in full responsibility on our part in accordance with the regulations of the university, faculty, and examination committee.

Ho Chi Minh city,
Ho Chi Minh City University of Technology
Nguyen Anh Lam
Dang Huu Nhat Hoang
Vu Minh Quan

ACKNOWLEDGEMENTS

First and foremost, we would like to express our sincere and profound gratitude to PhD. Le Thanh Sach for his dedicated supervision, invaluable guidance, and insightful feedback throughout the entire process of this Capstone Project. His professional expertise, academic rigor, and continuous encouragement have played a decisive role in shaping the direction, methodology, and quality of this work.

We would also like to extend our appreciation to the lecturers of the Faculty of Computer Science and Engineering for providing a solid theoretical foundation and practical knowledge during our academic studies, which have been essential for the successful completion of this project.

Our sincere thanks go to our classmates and friends for their constructive discussions, technical assistance, and moral support during the research and implementation phases of this project.

Finally, we would like to express our deepest gratitude to our families for their constant support, understanding, and encouragement throughout our academic journey. Their unconditional support has been a great source of motivation that enabled us to complete this graduation thesis.

Abstract

Recent advances in multimodal learning have created new opportunities for semantically informed analysis of 3D medical images. In this context, integrating caption-like clinical descriptions with volumetric visual features can improve anatomical localization and support more reliable recovery of clinically relevant regions.

In this Capstone Project, we investigate a multimodal framework for caption-conditioned anatomical localization on the RAOS dataset. The dataset is organized for 19 anatomical organs, including Liver, Spleen, Kidney (Left), Kidney (Right), Stomach, Gallbladder, Esophagus, Pancreas, Duodenum, Colon, Intestine, Adrenal (Left), Adrenal (Right), Rectum, Bladder, Head of femur (Left), Head of femur (Right), Prostate, and Seminal vesicles. The research process includes literature review, dataset analysis, model design, implementation, and experimental evaluation. Within the conceptual framing of this thesis, the textual input serves as a caption-like semantic query, and the resulting voxel mask can be interpreted as a retrieval-oriented localization output over 3D medical image space.

The proposed method combines a pretrained 3D Swin UNETR image encoder and a pretrained BioBERT text encoder through a cross-modal fusion module, followed by a 3D decoder for voxel-level prediction. In this pipeline, medical text descriptions provide caption-like semantic queries, while volumetric CT features preserve spatial anatomical context. To improve training efficiency and stability, the vision and language encoders are frozen, and only the fusion and decoder modules are optimized for downstream query-conditioned localization.

Experimental results indicate that this framework supports effective semantic target recovery in 3D medical images. These findings suggest that multimodal semantic alignment can enhance anatomically relevant region retrieval and provide a practical basis for retrieval-oriented localization in future clinical imaging systems.

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Chapter 1

INTRODUCTION

1.1 Motivation and Background

Recent progress in artificial intelligence, especially deep learning, has significantly improved computer vision systems for healthcare. In modern clinical workflows, 3D medical imaging modalities such as CT and MRI are essential because they provide detailed anatomical information for diagnosis, treatment planning, and follow-up assessment.

In practice, analyzing a 3D scan is labor-intensive. Clinicians often need to inspect many slices across axial, coronal, and sagittal planes, and final interpretation can depend on individual experience. As the number of studies increases, robust automated tools are increasingly necessary to support consistent and efficient medical image analysis.

Medical image segmentation is one of the most important tasks in this context because it localizes organs or lesions at voxel level and supports quantitative analysis and 3D visualization. However, many existing models rely only on visual features and do not explicitly exploit semantic signals that could help identify *which* anatomical structure should be recovered in a given image volume. This limitation becomes especially important when multiple organs share similar appearance or when boundaries are weak.

At the same time, multimodal learning has shown that language can provide useful prior knowledge for visual understanding. In medical imaging, caption-like textual cues can encode anatomical identity and contextual relationships that are difficult to infer from appearance alone. This motivates a caption-conditioned localization strategy in which text behaves as a semantic retrieval cue and the final voxel mask serves as a localization-oriented output rather than a standalone cap-

tion generation result.

Based on this motivation, this Capstone Project focuses on building a multimodal framework for query-conditioned anatomical localization on the RAOS dataset with 19 anatomical organs. The central idea is to fuse volumetric image features with medical semantic text representations in a unified model, thereby improving semantic target recovery and supporting retrieval-oriented localization for downstream clinical analysis.

1.2 Problem Statement

In this project, the research problem is to develop a multimodal framework for query-conditioned anatomical localization in three-dimensional medical imaging. Given a 3D medical image volume from the RAOS dataset and a caption-like anatomical description, the system must combine volumetric visual information with textual medical context in order to recover the anatomically relevant region as a voxel-level mask. Within the conceptual framing of this thesis, that mask is interpreted as a retrieval-oriented localization output for 19 anatomical organs, with text providing the semantic query signal.

More specifically, the challenge is to design a multimodal model that can encode 3D image features and medical text representations, fuse them effectively, and produce spatially consistent localization results across the full volume. The problem is complicated by the large size of 3D data, the variability of anatomical appearance, and the need to preserve spatial context while using text as a semantic query rather than merely an auxiliary label. Therefore, the core objective is to exploit caption-like semantic cues to improve the accuracy, robustness, and anatomical specificity of 3D region recovery.

1.3 Thesis Objectives

The primary objective of this project is to design and evaluate a multimodal framework for query-conditioned anatomical localization in three-dimensional medical images. The proposed system uses medical semantic text as a caption-like query signal to support anatomical localization and improve voxel-level recovery of clinically relevant regions.

The specific objectives of the project are as follows:

- To study the characteristics of the RAOS dataset and its 19 anatomical organs, and to prepare the image, label, and text data for caption-conditioned localization experiments.
- To implement a multimodal alignment pipeline that combines a pretrained 3D image encoder, a pretrained medical text encoder, a cross-modal fusion module, and a 3D decoder.
- To train the trainable components of the framework while keeping the pretrained encoders frozen, in order to improve training stability and computational efficiency.
- To evaluate the proposed model using standard segmentation metrics interpreted as measures of retrieval-oriented localization quality, and to visualize the recovered 3D anatomical regions for qualitative analysis.

Through these objectives, the project aims to demonstrate that integrating caption-like semantic queries with 3D visual features can support reliable anatomical localization and strengthen multimodal semantic alignment in medical imaging.

1.4 Scope and Limitations

Within the scope of this Capstone Project, the study is limited to the analysis of three-dimensional medical imaging data from the RAOS dataset, focusing on caption-conditioned anatomical localization. The proposed approach concentrates on recovering voxel-level anatomical regions for 19 organs by combining volumetric image features with caption-like semantic descriptions. The Capstone Project does not aim to perform clinical diagnosis, treatment planning, or medical decision-making.

The experiments are conducted using pretrained image and text encoders together with trainable fusion and decoding modules, instead of developing a completely new architecture from scratch. Performance evaluation is based on standard segmentation metrics and qualitative visualization of recovered 3D anatomical regions. Direct validation by medical experts and formal image-ranking retrieval benchmarks are outside the current scope of the project.

1.5 Research Contributions

This Capstone Project makes several key contributions to multimodal 3D medical image analysis:

- Proposed a multimodal framework for caption-conditioned anatomical localization on the RAOS dataset with 19 anatomical organs.
- Implemented and evaluated a Swin UNETR–BioBERT baseline with cross-modal attention to study semantic query alignment in 3D organ recovery.
- Designed and implemented a 3D localization pipeline with frozen encoders and trainable fusion and decoder components.
- Evaluated the proposed framework using standard segmentation metrics interpreted as measures of retrieval-oriented localization quality, together with qualitative 3D visualization.

1.6 Report Structure

This Capstone Project is organized as follows:

- **Chapter 1: Introduction** presents the motivation, problem statement, objectives, scope, contributions, and report structure.
- **Chapter 2: Literature Review** summarizes the background on deep learning, attention mechanisms, multimodal learning, and evaluation metrics.
- **Chapter 3: Related Works** reviews studies on medical image segmentation, text-guided segmentation, and referring expression segmentation.
- **Chapter 4: Proposed Method** describes the RAOS dataset, data preparation, and the proposed caption-conditioned localization architecture.
- **Chapter 5: Experimental Setup and Results** presents the experimental configuration, evaluation protocol, quantitative results, qualitative analysis, and discussion.
- **Chapter 6: Conclusion and Future Work** summarizes the project findings, limitations, and future research directions.

Chapter 2

LITERATURE REVIEW

2.1 Overview of Medical Image Segmentation

Medical image segmentation is the task of automatically identifying and isolating specific anatomical structures, organs, or pathological regions within medical images at the voxel or pixel level. In clinical practice, accurate segmentation is essential for diagnosis, treatment planning, surgery guidance, and longitudinal monitoring. Traditional manual segmentation by clinicians is labor-intensive, time-consuming, and subject to inter-observer variability.

With the rise of deep learning, convolutional neural networks (CNNs) and encoder-decoder architectures such as U-Net have demonstrated remarkable performance in 2D and 3D medical image segmentation tasks. However, many current approaches rely solely on visual features and do not explicitly leverage other available information, such as semantic descriptions of anatomical structures. This limitation motivates the integration of multimodal information, where textual semantics can provide additional context to support query-conditioned anatomical localization, especially in challenging cases with low contrast or ambiguous boundaries.

This chapter reviews the fundamental concepts and techniques relevant to the proposed multimodal framework. Although the downstream output remains a voxel-level mask, the discussion is organized to support the thesis framing in which caption-like semantic descriptions guide retrieval-oriented anatomical localization through deep multimodal alignment.

2.2 Deep Learning in Medical Imaging

2.2.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are deep neural network architectures tailored for image analysis and computer vision. Unlike fully connected networks, CNNs exploit spatial locality via learned convolutional filters, allowing automatic extraction of hierarchical features from raw pixel data without manual feature engineering.

The fundamental building block of a CNN is the convolution layer: a small kernel (filter) is convolved with the input to produce feature maps that emphasize image patterns such as edges, textures, and shapes. The 2D convolution operation can be written as

$$Y(i,j) = \sum_m \sum_n X(i+m, j+n)\, K(m,n) \tag{2.1}$$

where X denotes the input image, K the kernel, and Y the resulting feature map. After convolution, nonlinear activation functions (e.g., ReLU) and pooling layers are typically applied to increase representational power and reduce spatial resolution for computational efficiency.

Stacking multiple convolutional layers enables CNNs to learn hierarchical representations: early layers capture local, low-level patterns while deeper layers represent more complex, semantic structures. For volumetric medical images such as CT or MRI, convolutions can be extended to three dimensions (3D convolutions), which operate on cubic kernels and preserve inter-slice context at the cost of higher memory and compute requirements.

In medical image segmentation, CNN-based architectures (for example, U-Net) have become standard due to their ability to delineate organ boundaries and differentiate tissue types while retaining morphological details.

2.2.2 Encoder-Decoder Architectures

Encoder-decoder architectures form a fundamental paradigm in semantic segmentation and dense prediction tasks. These networks consist of two complementary components: an encoder that progressively compresses the input to capture high-level semantic information, and a decoder that reconstructs spatial details to produce a dense pixel-level or voxel-level prediction map. This design elegantly balances the need for global context with the preservation of fine-grained spatial

resolution.

U-Net, proposed by Ronneberger et al., is a landmark encoder-decoder network specifically designed for medical image segmentation with limited training data. The architecture features a symmetric structure: the encoder arm performs sequential convolutions and max pooling to downsample the feature maps, while the decoder arm mirrors this structure through upsampling (via transposed convolutions or bilinear interpolation) and convolution. The defining characteristic of U-Net is its skip connections: feature maps from each level of the encoder are concatenated with the corresponding decoder feature maps at the same resolution. These skip connections enable the network to combine localized spatial information captured early in the encoding path with semantic context learned at deeper layers, thereby preserving fine morphological details essential for accurate segmentation.

Mathematically, if the encoder produces a sequence of feature maps at resolutions $\{f_1, f_2, \ldots, f_d\}$ and the decoder generates upsampled maps $\{g_d, g_{d-1}, \ldots, g_1\}$, the skip connection concatenates them as $\hat{g}_i = \text{concat}(g_i, f_i)$, which is then processed by convolutional layers. This mechanism has proven particularly effective in medical imaging, where segmentation boundaries must be crisp and detailed.

U-Net and its variants (such as 3D U-Net for volumetric data) have become standard backbones in medical image segmentation. Extensions of this architecture include incorporation of batch normalization, residual connections, and attention mechanisms to further improve performance and training stability.

2.2.3 Sliding Window Inference

The sliding window approach is a practical inference strategy for processing large 3D medical images that exceed GPU memory capacity. Rather than processing the entire volumetric data in a single forward pass, the method divides the 3D volume into smaller, overlapping cubic or rectangular patches (or windows). Each patch is processed independently through the network, and predictions are aggregated to reconstruct a full-resolution segmentation map of the original volume.

Mathematically, if a volume $V \in \mathbb{R}^{D \times H \times W}$ represents the full 3D image and a window size is $(d \times h \times w)$, the sliding window generates overlapping patches by sliding the window with stride (s_d, s_h, s_w). For each patch P_i, the network computes a corresponding segmentation prediction $\hat{S}_i$. Overlapping predictions are then combined, typically via averaging or weighted voting, to produce the final

segmentation S_{full} for the entire volume.

The overlap strategy is crucial: windows that are too small or non-overlapping can lead to boundary artifacts at patch edges, whereas sufficient overlap (typically 50% or more along each dimension) allows smooth blending of predictions. This approach enables inference on volumes that would otherwise require prohibitive GPU memory, making it indispensable for clinical 3D CT and MRI applications where volumes may exceed 400×400×400 voxels.

Sliding window inference trades computational efficiency (memory savings) for increased computation time, since overlapping patches lead to redundant processing of boundary regions. However, this trade-off is often acceptable in clinical practice, where inference speed is less critical than memory efficiency and segmentation accuracy.

2.3 Attention Mechanisms and Transformers

2.3.1 Scaled Dot-Product Attention

Scaled Dot-Product Attention is the core of Transformer mechanism, used to measure the relevance of each element in a sequence through three matrices: Query matrix (Q), Key matrix (K), Value matrix (V). The attention logits are scaled by $\frac{1}{\sqrt{d_k}}$ so that their magnitude remains well behaved as the key dimension increases, which stabilizes optimization. This helps the training process more stable. After that, the result is passed through the softmax function to generate the attention weight. This approach allows the model focus on most important parts of the input data in an efficient way.

The whole process is summarized as shown in the following figure:

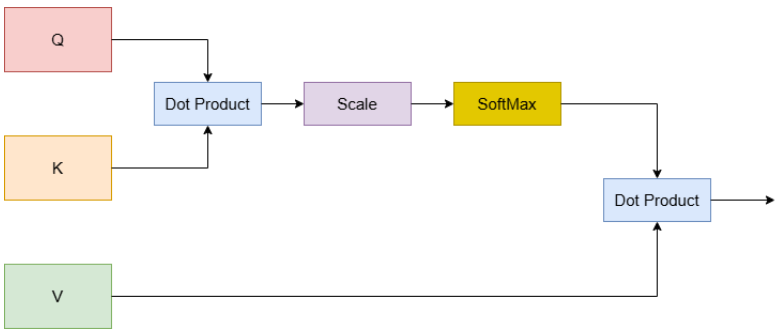


Figure 2.1: Scaled Dot-Product Attention

And it is summarized by the following formula:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_K}})V \text{ [5]}$$

2.3.2 Self-Attention

Self-Attention is a special case of attention in which Query matrix, Key matrix and Value matrix are all generated from the same input. This mechanism allows each element in the sequence to learn the relationship with all the other elements in that sequence, regardless of the distance between them. Therefore, it is better to model long-term dependencies than traditional sequential models. Self-Attention plays a important role, it completely eliminating sequential operations and allowing parallel processing of the whole input sequence.

2.3.3 Multi-Head Self-Attention

Multi-Head Self-Attention expands Self-Attention by performing many attention processes in parallel by different sub-performance spaces. Instead of using a single attention, the model projects Query, Key, and Value into many subsets with different weights, each corresponding to a "Head". Each Head learns different type of relationship in the data, such as semantic, syntactic, or positional. The results from the Heads are concatenated and projected linearly to produce the final representation, which increases the expressiveness of the model without increasing the computational cost too much.

Residual Connection is used in Transformer to improve gradient propagation and help train deep models more effectively. In each Transformer block, the input of a sublayer is added directly to the output of that sublayer before moving on to the next step. This mechanism allows the model to preserve the original information and reduce vanishing gradient.

2.3.4 Layer Normalization

Layer Normalization is a normalization technique applied in Transformer to stabilize and speed up the training process. Layer Normalization normalizes the feature-oriented characteristics of each individual sample. Layer Normalization is commonly applied after adding residual connections, which ensures that the output distribution of each block is always stable, thereby improving the convergence of the model.

2.3.5 Feed-Forward Network

The Feed-Forward Network in Transformer is a fully connected layer with two networks that is applied independently and identically to each location. It consists of a characteristic spatial extension linear layer, followed by a nonlinear activation function, and a second linear layer to bring the representation back to the original dimension. The Feed-Forward Network plays an important role in increasing the nonlinear representation of the model after the attention process.

2.3.6 Transformer Encoder Block

The Transformer Encoder Block is the basic structural unit of encoders in Transformer, consisting of two main subclasses: Multi-Head Self-Attention and Feed-Forward Network. Each sub-layer is bounded by Residual Connection and Layer Normalization. This structure allows the model to learn the relationship through attention and enhance the typical representation at each position through Feed-Forward Network. Stacking many encoder blocks helps Transformer build more abstract and more information representations.

2.4 Evaluation Metrics

2.4.1 Segmentation Metrics (Dice Score, HD95)

In medical image segmentation, evaluation metrics quantify the quality of predicted segmentation masks against ground truth annotations. Two widely-used metrics are the Dice Similarity Coefficient (DSC) and the Hausdorff Distance at the 95th percentile (HD95).

1. **Dice Similarity Coefficient (Dice Score)**

   The Dice Similarity Coefficient measures the spatial overlap between predicted and ground truth regions, ranging from 0 to 1, where 1 indicates perfect agreement. For a predicted segmentation mask P and ground truth mask G, the Dice score is defined as:

   $$\text{Dice}(P, G) = \frac{2 \times |P \cap G|}{|P| + |G|} \tag{2.2}$$

   where $|P \cap G|$ is the cardinality of the intersection and $|P|$, $|G|$ are the cardinalities of the predicted and ground truth sets, respectively. The Dice score is particularly useful for segmentation tasks because it emphasizes the overlap region rather than penalizing minor boundary variations heavily.

2. **Hausdorff Distance at 95th Percentile (HD95)**

   The Hausdorff Distance (HD) measures the boundary distance between predicted and ground truth segmentations. Since the standard Hausdorff Distance is sensitive to outliers caused by a few misclassified voxels, HD95 uses the 95th percentile of surface distances to provide a more robust evaluation:

$$\text{HD95}(P, G) = \max\left\{\text{percentile}_{95}(d(P, G)), \text{percentile}_{95}(d(G, P))\right\} \tag{2.3}$$

   where

$$d(P, G) = \left\{ \min_{g \in \partial G} \|p - g\| \;\middle|\; p \in \partial P \right\} \tag{2.4}$$

   and ∂P and ∂G denote the boundaries of the predicted and ground truth regions, respectively. HD95 is typically expressed in millimeters (or voxel units), and lower values indicate better boundary alignment accuracy.

Together, Dice and HD95 provide complementary views: Dice measures volumetric overlap, while HD95 emphasizes boundary precision, making them standard metrics in medical image segmentation benchmarks.

2.4.2 Training Loss Functions

During model training, loss functions guide the optimization process by providing numerical feedback on segmentation quality. The Generalized Dice Loss (GDL) is a widely-used loss function for multi-class segmentation that addresses class imbalance and encourages focus on clinically important structures.

The Dice Loss (also called Soft Dice Loss) is the complement of the Dice score and is directly optimizable via backpropagation:

$$\mathcal{L}_{\text{Dice}} = 1 - \text{Dice}(P, G) = 1 - \frac{2 \times |P \cap G|}{|P| + |G|} \tag{2.5}$$

The Generalized Dice Loss extends this to multi-class scenarios by weighting each class according to its inverse frequency:

$$\mathcal{L}_{\text{GDL}} = 1 - \frac{2 \sum_c w_c |P_c \cap G_c|}{\sum_c w_c (|P_c| + |G_c|)} \tag{2.6}$$

where $w_c = \frac{1}{f_c}$ is the weight for class c, and f_c is the frequency (proportion of ground truth voxels) of class c in the training data. By down-weighting frequently-occurring classes (e.g., background), GDL prevents the model from ignoring rare anatomical structures.

Cross-entropy loss is another common choice, but it can be insensitive to class imbalance. In practice, many implementations use hybrid losses combining Dice Loss and Cross-Entropy to balance boundary precision and probabilistic calibration. The choice of loss function depends on dataset characteristics and task priorities (volumetric accuracy vs boundary precision).

Chapter 3

RELATED WORKS

3.1 Overview

This chapter provides a comprehensive review of the key research directions that underpin the proposed framework. The literature is organized into three thematic areas, each addressing a critical component of the system architecture.

First, Section 3.2 examines state-of-the-art **3D medical image segmentation architectures**, tracing the evolution from convolutional encoder–decoder networks such as 3D U-Net [6] to Transformer-based approaches exemplified by Swin UNETR [4] and SAM-Med3D [3]. These works establish the visual backbone upon which our image encoder is built.

Second, Section 3.3 surveys **language-conditioned anatomical localization** methods, focusing on how natural language prompts can act as semantic queries that steer localization toward a specific anatomical region of interest. This section reviews BioBERT [1] as a domain-specific language encoder and discusses the cross-modal attention mechanism [5] that enables image–text alignment.

Third, Section 3.4 introduces **Referring Expression Segmentation (RES)**—the task paradigm most closely aligned with the core idea of this project. We review the Multi-Granularity RES (MRES) framework proposed by Wang et al. [2], which serves as the primary inspiration for our approach: using a free-form text query to *retrieve* a target anatomical structure within a 3D CT volume through localization.

Together, these three pillars—volumetric localization, multimodal semantic querying, and referring expression grounding—form the theoretical and methodological foundation of the proposed framework.

3.2 3D Medical Image Segmentation Architectures

Medical image segmentation in three dimensions poses unique challenges compared to its two-dimensional counterpart. Volumetric data, such as CT and MRI scans, contain rich spatial context along the axial, coronal, and sagittal planes, yet this comes at the cost of substantially higher computational demands and memory consumption. This section reviews the key architectural milestones that have shaped the field and directly inform the design of the image encoder in our proposed framework.

3.2.1 3D U-Net

Çiçek et al. [6] extended the seminal U-Net architecture [7] from 2D to 3D by replacing all two-dimensional operations—convolution, max-pooling, and up-convolution—with their volumetric equivalents. The resulting network follows a symmetric encoder–decoder topology:

- **Encoder (contraction path):** A sequence of $3 \times 3 \times 3$ convolutional blocks, each followed by $2 \times 2 \times 2$ max-pooling, progressively reduces the spatial resolution while increasing the number of feature channels.
- **Decoder (expansion path):** Symmetric up-convolution layers restore the spatial resolution. At each level, feature maps from the corresponding encoder stage are concatenated via *skip connections*, allowing the decoder to recover fine-grained spatial details that would otherwise be lost during downsampling.

The skip connection mechanism is of particular importance for medical image segmentation, where organ boundaries are often subtle and low-contrast. Without these connections, the decoder would rely solely on the highly compressed bottleneck representation, leading to imprecise and blurry segmentation masks.[1]

Furthermore, 3D U-Net demonstrated that dense volumetric segmentation could be learned from *sparsely annotated* data—a property with practical relevance in clinical settings where expert annotation of every slice is prohibitively expensive. This work established the foundational design pattern that all subsequent 3D segmentation architectures, including the one employed in this project, build upon.

[1] In our implementation, the decoder (`Decoder3D`) explicitly concatenates encoder skip features at each upsampling stage via the `UpSkipBlock3D` module.

3.2.2 SAM-Med3D

The Segment Anything Model (SAM) [8] demonstrated remarkable zero-shot segmentation capabilities on natural images through a prompt-driven paradigm. SAM-Med3D [3] adapts this paradigm for volumetric medical data by extending all processing blocks from 2D to 3D. The architecture comprises three principal components:

1. **3D Image Encoder:** Divides the input volume into non-overlapping 3D patches, maps them to an embedding space, and processes them through a sequence of Transformer encoder blocks with 3D self-attention. This mechanism enables the model to capture long-range spatial dependencies across the entire volume.

2. **3D Prompt Encoder:** Encodes user-provided prompts—such as point coordinates, bounding boxes, or coarse masks—into embeddings that are dimensionally compatible with the image encoder output. This component enables interactive, prompt-guided segmentation.

3. **3D Mask Decoder:** Generates the final segmentation mask by attending to both the image embeddings and the prompt embeddings through Transformer cross-attention layers.

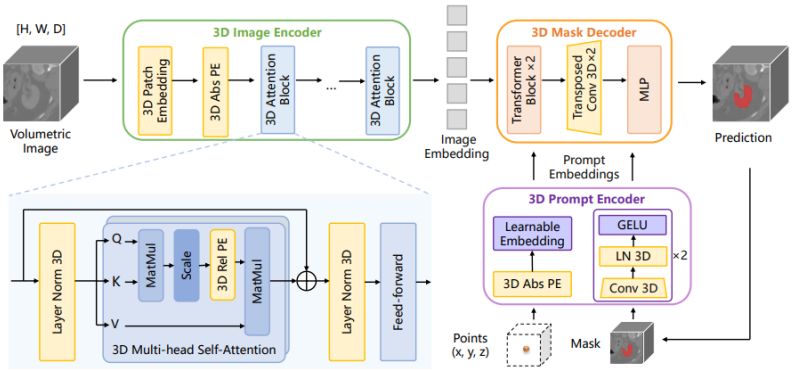


Figure 3.1: Architecture of SAM-Med3D. The model extends SAM's three-component design (image encoder, prompt encoder, mask decoder) from 2D to 3D, enabling volumetric medical image segmentation guided by spatial prompts. Adapted from [3].

SAM-Med3D's prompt-driven design is conceptually related to our framework's use of text prompts to guide segmentation. However, while SAM-Med3D relies on *spatial* prompts (points, boxes), our approach employs *semantic* prompts expressed in natural language, requiring a fundamentally different fusion mechanism.

3.2.3 Swin UNETR

Hatamizadeh et al. [4] proposed Swin UNETR (Swin UNEt TRansformers), which replaces the convolutional encoder of the U-Net architecture with a Swin Transformer [9] backbone, while retaining a CNN-based decoder with skip connections. This hybrid design combines the long-range modeling capability of Transformers with the spatial precision of convolutional decoders.

3.2.3.1 Hierarchical Swin Transformer Encoder

The encoder operates as follows. The input 3D volume $\mathbf{X} \in \mathbb{R}^{1\times D\times H\times W}$ is first partitioned into non-overlapping 3D patches and linearly embedded into a sequence of tokens. These tokens are then processed through four stages of Swin Transformer blocks, each consisting of alternating layers of *Window-based Multi-Head Self-Attention* (W-MSA) and *Shifted Window Multi-Head Self-Attention* (SW-MSA):

- **W-MSA:** Self-attention is computed within local, non-overlapping 3D windows of size $M \times M \times M$, reducing the computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \cdot M^3)$, where N is the total number of tokens.
- **SW-MSA:** The window partition is shifted by $\lfloor M/2 \rfloor$ voxels in each dimension, enabling cross-window information flow without the overhead of global self-attention.

Patch merging layers between stages downsample the spatial resolution by a factor of 2 while doubling the feature dimension, producing a multi-scale feature hierarchy at four resolution levels.[2]

3.2.3.2 Shifted Window Mechanism

The shifted window strategy is a key innovation that distinguishes Swin Transformer from the original Vision Transformer (ViT) [10]. By alternating between regular and shifted window partitions across consecutive Transformer layers, the network achieves cross-window connectivity while maintaining linear computational complexity with respect to input size. This is particularly advantageous for 3D medical volumes, where global self-attention over the entire volume would be computationally prohibitive.

[2]In our implementation, we use a Swin UNETR encoder with `feature_size=48`, `depths=(2,2,2,2)`, `num_heads=(3,6,12,24)`, and `window_size=7`, yielding skip features with channel dimensions of 48, 96, 192, and 384.

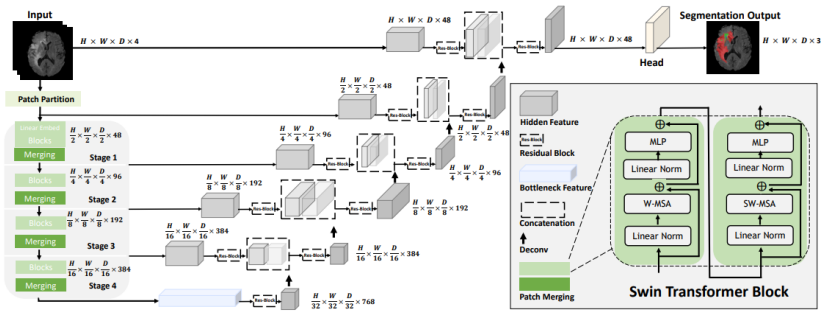


Figure 3.2: Overview of the Swin UNETR architecture. The Swin Transformer encoder extracts hierarchical features at four resolution levels, connected to a CNN-based decoder via skip connections. Adapted from [4].

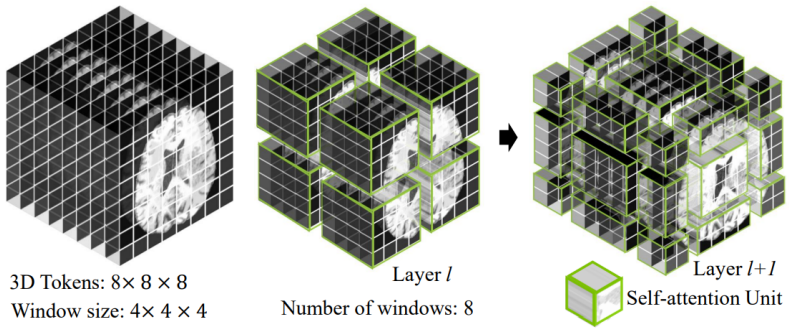


Figure 3.3: Illustration of the shifted windowing mechanism. In Layer l, attention is computed within regular windows. In Layer $l + 1$, the window grid is shifted by half the window size, enabling cross-window information exchange. Adapted from [4].

3.2.3.3 Relevance to Our Framework

Swin UNETR serves as the **image encoder backbone** in our proposed multimodal localization framework. Specifically, we extract the `swinViT` component of the pre-trained Swin UNETR and repurpose it as a 3D feature extractor. The bottleneck feature map (at the lowest spatial resolution) is reshaped into a token sequence and projected to a shared embedding dimension of $d = 768$, enabling subsequent cross-modal fusion with text features. The four intermediate skip features are passed directly to the decoder to preserve fine-grained spatial information for voxel-level region recovery.[3]

[3]This decomposition of Swin UNETR into a standalone encoder is implemented in `ImageEncoder3D`, which accesses the `.swinViT` sub-module and adds a linear projection head.

3.2.4 Summary and Comparison

Table 3.1 provides a comparative summary of the reviewed segmentation architectures.

Table 3.1: Comparison of 3D medical image segmentation architectures.

Model	Encoder Type	Prompt Support	Long-Range Modeling
3D U-Net [6]	CNN	None	Limited by kernel size
SAM-Med3D [3]	ViT (3D)	Spatial	Global self-attention
Swin UNETR [4]	Swin (3D)	None*	Shifted window attention

*In its original formulation. Our framework extends Swin UNETR with a text-based prompt mechanism via cross-modal attention.

As shown in the table, while CNN-based architectures (3D U-Net, V-Net) laid the structural foundation with skip connections and the Dice Loss, they are inherently limited in capturing long-range dependencies. SAM-Med3D addresses this with full 3D self-attention and prompt support, but only accepts spatial prompts. Swin UNETR offers an efficient compromise through shifted window attention. Our framework builds upon Swin UNETR's encoder and extends it with *semantic* (text-based) prompt support through the cross-modal fusion mechanism described in the next section.

3.3 Caption-Conditioned Anatomical Localization

Traditional medical image segmentation operates in a purely visual paradigm: a model receives an image (or volume) as input and produces a segmentation mask as output, without any mechanism for specifying *which* structure should be recovered at inference time. This limitation necessitates either training a separate model for each target organ or employing a multi-class output head with a fixed set of anatomical labels. Neither approach generalizes naturally to open-vocabulary semantic queries such as "segment the left kidney" or "highlight the spleen."

A caption-conditioned localization formulation addresses this limitation by introducing a natural language prompt as an additional input modality. In this setting, the prompt functions as a caption-like semantic query, and the model must jointly process the visual content and the textual description to recover the anatomically relevant region. The resulting mask can therefore be interpreted as a retrieval-oriented localization output rather than only as a conventional segmentation map. This paradigm requires two key capabilities: (1) a language encoder

that can represent medical terminology in a semantically meaningful embedding space, and (2) a fusion mechanism that enables the text representation to modulate visual feature extraction.

3.3.1 BioBERT: Domain-Specific Language Encoding

The choice of language encoder is critical for query-conditioned anatomical localization. General-purpose models such as BERT[4] are pre-trained on corpora dominated by general-domain text (Wikipedia, BookCorpus), and consequently underperform on biomedical terminology due to the distributional shift between general and medical language.

Lee et al. [1] addressed this gap by introducing BioBERT—a domain-specific language representation model that initializes from the BERT-base architecture (12 Transformer layers, 768 hidden units, 12 attention heads) and continues pre-training on large-scale biomedical corpora:

- **PubMed Abstracts:** Approximately 4.5 billion words from biomedical literature abstracts.
- **PMC Full-Text Articles:** Approximately 13.5 billion words from open-access biomedical publications.

This additional pre-training enables BioBERT to construct contextually rich representations for medical terms that are rare or absent in general corpora. For example, terms such as "hepatomegaly," "renal cortex," or "splenomegaly" receive meaningful embeddings that capture their anatomical and pathological semantics.[5]

[4]Bidirectional Encoder Representations from Transformers (BERT) [11].

[5]In our implementation, we use `dmis-lab/biobert-base-cased-v1.1` as the pre-trained checkpoint for the text encoder.

BioBERT demonstrated significant improvements over BERT on three representative biomedical NLP benchmarks:

Table 3.2: Performance comparison between BERT and BioBERT on biomedical NLP tasks, as reported in [1].

Task	**BERT**	**BioBERT**
Named Entity Recognition (F1)	85.76%	86.38% (+0.62)
Relation Extraction (F1)	72.28%	75.08% (+2.80)
Question Answering (MRR)	24.46%	36.70% (+12.24)

In our framework, BioBERT processes the input text prompt and produces a sequence of contextualized token embeddings $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_L] \in \mathbb{R}^{L \times d_{\text{text}}}$, where L is the number of tokens and $d_{\text{text}} = 768$. These embeddings are then projected through a learnable linear layer to match the dimension of the image features:

$$\mathbf{T}' = \mathbf{T} \cdot \mathbf{W}_{\text{proj}} + \mathbf{b}_{\text{proj}}, \quad \mathbf{W}_{\text{proj}} \in \mathbb{R}^{d_{\text{text}} \times d} \tag{3.1}$$

where d is the shared embedding dimension used for cross-modal fusion.

3.3.2 Cross-Modal Attention for Image–Text Fusion

The core mechanism enabling this caption-conditioned localization behavior is *cross-modal attention*, a variant of the Multi-Head Attention mechanism introduced in the Transformer architecture [5]. Unlike self-attention, where Query, Key, and Value all derive from the same modality, cross-modal attention draws the Query from one modality and the Key–Value pair from another.

In our framework, the cross-modal attention is configured as follows:

$$\mathbf{Q} = \mathbf{X}_{\text{img}}, \quad \mathbf{K} = \mathbf{V} = \mathbf{T}' \tag{3.2}$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{3.3}$$

where $\mathbf{X}_{\text{img}} \in \mathbb{R}^{N \times d}$ is the image feature sequence from the Swin Transformer encoder ($N = D' \times H' \times W'$ spatial tokens) and $\mathbf{T}' \in \mathbb{R}^{L \times d}$ is the projected text feature sequence.[6]

[6]This corresponds to the `CrossModalFusion` module in our codebase, which uses `nn.MultiheadAttention` with 8 attention heads.

The attention output is integrated with the original image features via a residual connection and Layer Normalization:

$$\mathbf{X}_{\text{fused}} = \text{LayerNorm}(\mathbf{X}_{\text{img}} + \text{CrossAttn}(\mathbf{X}_{\text{img}}, \mathbf{T}', \mathbf{T}')) \tag{3.4}$$

This formulation has several important properties:

1. **Selective attention:** Each image token computes attention weights over all text tokens, effectively learning which words in the prompt are most relevant to its spatial location. For example, when processing a CT volume with the prompt "liver," image tokens corresponding to the hepatic region will attend strongly to the token embedding of "liver."

2. **Information preservation:** The residual connection $\mathbf{X}_{\text{img}} + (\cdot)$ ensures that the original visual information is preserved, with the text-derived signal serving as a *complementary modulation* rather than a replacement.

3. **Multi-head decomposition:** With $h = 8$ attention heads, each operating on a $d/h = 96$-dimensional subspace, the model can simultaneously capture different aspects of the image–text relationship (e.g., semantic correspondence, spatial localization, anatomical context).

This cross-modal fusion mechanism transforms a generic 3D segmentation backbone into a semantically queryable localization system: the same model can recover different anatomical regions by simply changing the input text prompt, without retraining or modifying the network architecture.

3.4 Referring Expression Segmentation

The preceding sections reviewed the architectural building blocks for 3D segmentation (Section 3.2) and the mechanisms for incorporating textual guidance (Section 3.3). This section introduces *Referring Expression Segmentation* (RES)—the task paradigm that conceptually unifies these components and most directly aligns with the objective of this thesis.

3.4.1 Task Definition and Motivation

Referring Expression Segmentation (RES) is defined as follows: given an image $\mathbf{I}$ and a natural language expression $\mathcal{E}$, the task is to produce a binary segmentation mask $\mathbf{M}$ that delineates the region in $\mathbf{I}$ described by $\mathcal{E}$:

$$f : (\mathbf{I}, \mathcal{E}) \longrightarrow \mathbf{M} \in \{0, 1\}^{D \times H \times W} \tag{3.5}$$

Unlike traditional semantic segmentation, which assigns a pre-defined class label to every pixel from a closed vocabulary, RES requires the model to *understand* the meaning of a free-form text query and *locate* the corresponding region in the image. This distinction is fundamental: semantic segmentation answers "*what is everywhere?*" while RES answers "*where is this specific thing?*"

Crucially, RES establishes a natural bridge between *image retrieval* and *image segmentation*. The text query functions as a retrieval query, and the segmentation mask functions as the retrieval result—the model "retrieves" the target entity from the image by segmenting it. This interpretation directly motivates the title and core contribution of this thesis: *Analyzing the Role of Image Captioning in Enhancing Text-to-Image Retrieval Performance.*

3.4.2 Multi-Granularity Referring Expression Segmentation

Wang et al. [2] advanced the RES paradigm by proposing the *Multi-Granularity Referring Expression Segmentation* (MRES) task, which extends classic object-level RES to fine-grained *part-level* grounding. Their work makes three principal contributions:

3.4.2.1 The MRES Task

While prior RES datasets and methods assumed that each expression refers to a complete object, MRES relaxes this assumption to include part-level targets. For example, an expression such as "the left eye of the cat" requires the model to segment a specific *part* of an object, demanding finer-grained visual–linguistic understanding. This multi-granularity capability is highly relevant to medical imaging, where clinicians routinely distinguish between anatomical substructures (e.g., "left kidney" vs. "right kidney," or "liver" vs. "left lobe of the liver").

3.4.2.2 The MRES-32M Dataset

To support the MRES task, the authors developed an automatic, model-assisted data engine to construct **MRES-32M**—a large-scale visual grounding dataset comprising over 32.2 million high-quality masks with corresponding natural language captions across 1 million images. This dataset is orders of magnitude larger than existing RES benchmarks such as RefCOCO, RefCOCO+, and RefCOCOg, enabling training of more robust and generalizable grounding models.

3.4.2.3 The UniRES Model

The authors proposed **UniRES**, a unified model for both object-level and part-level referring expression segmentation. The architecture follows a *dual-encoder, fusion, decoder* paradigm:

1. **Vision Encoder:** Extracts multi-scale visual features from the input image.
2. **Language Encoder:** Processes the text expression into a sequence of token embeddings.
3. **Cross-Modal Fusion:** Aligns and integrates visual and linguistic representations through attention-based mechanisms.
4. **Mask Decoder:** Generates the final segmentation mask from the fused multimodal features.

This four-component pipeline is directly mirrored in our proposed framework, as summarized in Table 3.3.

Table 3.3: Architectural correspondence between UniRES [2] and our proposed framework.

Component	**UniRES (2D Natural)**	**Ours (3D Medical)**
Vision Encoder	2D Vision Backbone	Swin ViT 3D [4]
Language Encoder	General Text Encoder	BioBERT [1]
Fusion Module	Cross-Modal Attention	Multi-Head Cross-Attention
Mask Decoder	2D Mask Head	3D U-Net Decoder [6]
Input Domain	Natural images (2D)	Abdominal CT volumes (3D)
Granularity	Object + Part	Organ-level
Eval. Metrics	cIoU, gIoU	Dice, HD95

3.4.3 Adaptation to 3D Medical Imaging

While the MRES framework operates on 2D natural images, our project adapts the core RES paradigm to the domain of **3D abdominal CT imaging**. This adaptation involves several non-trivial modifications:

1. **3D Vision Encoder:** The 2D vision backbone of UniRES is replaced with a 3D Swin Transformer encoder (extracted from Swin UNETR [4]), capable of processing volumetric CT data and capturing spatial relationships along all three anatomical axes.

2. **Domain-Specific Language Encoder:** Instead of a general-purpose text encoder, we employ BioBERT [1], which is pre-trained on biomedical corpora and thus better equipped to represent medical terminology such as organ names, anatomical landmarks, and pathological descriptions.

3. **Volumetric Decoder with Skip Connections:** The 2D mask head is replaced with a 3D decoder following the U-Net paradigm [6], incorporating four levels of skip connections from the Swin encoder to preserve fine-grained spatial details for accurate organ boundary delineation.

4. **Medical Evaluation Protocol:** The evaluation metrics are adapted from the computer vision domain (cIoU, gIoU) to the medical imaging standard: *Dice Score* [12] for volumetric overlap and *95th percentile Hausdorff Distance* (HD95) for boundary accuracy.

In essence, our system operates as follows: given a 3D CT volume $\mathbf{V} \in \mathbb{R}^{1\times D\times H\times W}$ and a natural language prompt $\mathcal{P}$ (e.g., "liver" or "left kidney"), the model produces a volumetric segmentation mask $\mathbf{M} \in \{0,1\}^{D\times H\times W}$ that delineates the described organ. From the user's perspective, this achieves *text-to-organ retrieval*—the system "finds" and visually isolates the queried anatomical structure within the CT scan.

3.5 Summary

This chapter has reviewed the three research pillars upon which our framework is built:

1. **3D Segmentation Architectures** (Section 3.2): From the foundational skip connections of 3D U-Net [6] and the Dice Loss of V-Net [12], through the prompt-driven paradigm of SAM-Med3D [3], to the efficient Transformer-based encoding of Swin UNETR [4].

2. **Text-Guided Segmentation** (Section 3.3): The use of BioBERT [1] for domain-specific language encoding and cross-modal attention [5] for image–text feature fusion.

3. **Referring Expression Segmentation** (Section 3.4): The MRES paradigm [2], which frames segmentation as a form of visual retrieval guided by natural language, and its adaptation to the 3D medical domain.

The next chapter describes the datasets used for training and evaluation, followed by the detailed methodology of the proposed multimodal framework.

Chapter 4

PROPOSED METHOD

4.1 Pipeline Overview

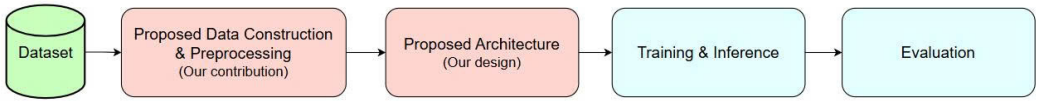


Figure 4.1: Overview of the pipeline.

Figure 4.1 illustrates the overall workflow of the proposed method. The framework consists of five major stages: dataset preparation, proposed data construction and preprocessing, proposed multimodal architecture, training and inference, and evaluation.

The pipeline begins with the dataset, which provides abdominal CT volumes together with anatomical organ annotations. Based on this raw dataset, our team proposed and implemented a dedicated data construction and preprocessing pipeline specifically designed for multimodal organ segmentation. This stage reorganizes the original dataset into paired image–label–text samples suitable for text-guided learning. In addition, preprocessing operations such as organ-specific mask generation, text prompt construction, spatial normalization, intensity preprocessing, and patch sampling are applied to standardize the training data and improve robustness.

After data preparation, our team designed a multimodal segmentation architecture that integrates both visual and textual information. The proposed archi-

tecture combines a SwinUNETR-based image encoder, a BioBERT text encoder, a cross-modal fusion module, and a 3D decoder to perform caption-conditioned organ segmentation from volumetric CT scans.

The processed data is then used during the training and inference stages, where the model learns to generate voxel-level segmentation masks from image–text inputs. Finally, the performance of the proposed framework is evaluated using quantitative metrics and qualitative visualization analysis.

4.2 Dataset and Preparation

4.2.1 Introduction to the RAOS Dataset

4.2.1.1 Overview and Objectives

RAOS (Rethinking Abdominal Organ Segmentation) is a thoracic–abdominal medical imaging dataset specifically designed for robustness evaluation of multi-organ segmentation models in real clinical scenarios. The primary objective of RAOS is to provide a comprehensive benchmark for evaluating the robustness of abdominal organ segmentation models in difficult clinical scenarios, such as organ deformation, partial organ loss, and complete organ removal.

Unlike conventional segmentation datasets that typically contain only healthy cases with intact anatomies, RAOS addresses a critical gap in medical imaging research by incorporating diverse pathological conditions commonly encountered in clinical practice. This dataset recognizes that real-world applications of organ segmentation models must handle surgical interventions, anatomical variations, and disease-induced organ modifications. By providing a diverse collection of clinical scenarios, RAOS enables researchers to develop and evaluate segmentation algorithms that are robust and generalizable to varied patient populations and clinical conditions.

4.2.1.2 Size and Content

The RAOS dataset consists of 413 real clinical 3D CT scans, each corresponding to a single patient. In addition, for each CT scan, nine synthesized MR scans are provided. A total of 19 abdominal and pelvic organs are annotated by a senior oncologist with over 10 years of clinical experience.

The dataset provides a multi-modal imaging perspective with both real CT scans and synthesized MR images. The comprehensive annotation of 19 organs including major structures such as the liver, kidneys, pancreas, spleen, stomach, and various blood vessels has been carefully validated by an experienced oncologist to ensure anatomical accuracy and clinical relevance. This raw multi-modal and

multi-organ data requires appropriate preprocessing and normalization before being utilized for model training, ensuring consistent image quality and standardized organ label representations.

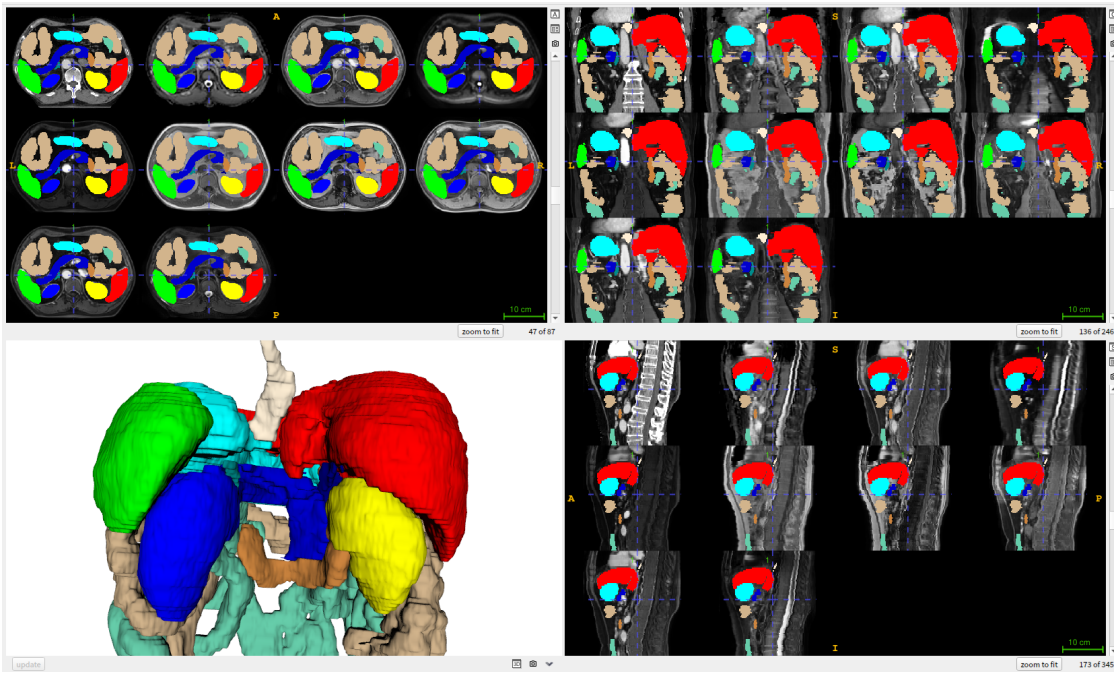


Figure 4.2: An example in the RAOS dataset (one CT and nine synthesized MR scans).

4.2.1.3 Clinical Subsets

To reflect the diversity of real clinical cases, RAOS is divided into three subsets:

- **Set A**: 317 CT volumes with intact abdominal organs. This subset represents baseline cases with normal anatomy, providing raw training data for typical clinical presentations.
- **Set B**: 22 CT volumes with partially removed tissues. This subset includes cases where surgical interventions or pathological conditions have resulted in partial organ removal or tissue loss, providing diverse training data with anatomical variations.
- **Set C**: 74 CT volumes with at least one organ completely removed. This subset represents challenging edge cases where complete organ removal or absence has occurred, providing diverse data for preprocessing and training.

The stratified design of RAOS with these three clinical subsets provides diverse raw medical imaging data for preprocessing and model training. Before utilizing RAOS for model development, comprehensive data preprocessing steps are required to handle the varying anatomical conditions, normalize imaging characteristics, handle missing organs, and prepare the data in a format suitable for deep learning algorithms. These preprocessing procedures ensure that the diverse clinical scenarios within RAOS can be effectively prepared and leveraged for training robust segmentation models.

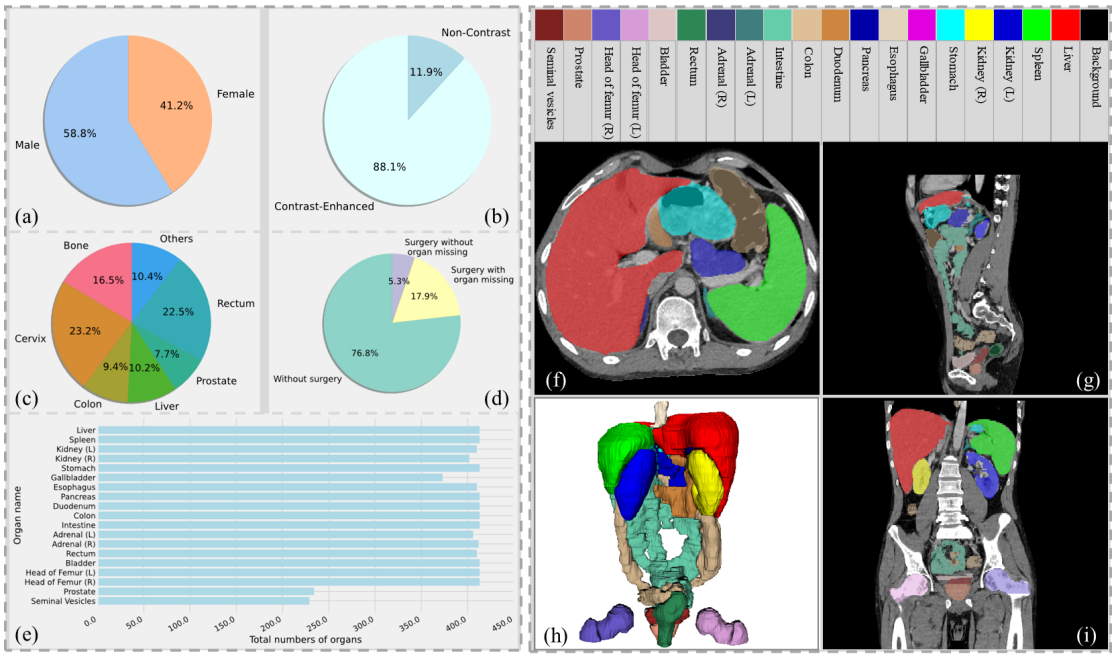


Figure 4.3: Clinical distribution of the RAOS dataset.

4.2.2 Dataset Creation and Processing Pipeline

4.2.2.1 Multi-modal Data Pairing (Text-Label-Image)

In RAOS dataset, each sample is represented as:

- **Image**: A 3D CT scan volume.
- **Label**: A segmentation mask that includes all organs, where each organ is assigned a unique ID.

Therefore, we decided to reorganize each sample as follows:

4.2.2.1.a Image and Label Representation

- **Image**: A volumetric CT scan block stored in NIfTI format. Each image is a 3D volume.
- **Label**: A binary segmentation mask that isolates one organ from the CT volume. Although the original dataset may contain multiple organs with distinct IDs, we convert each sample into a single-organ mask by filtering the original label according to organ-specific IDs.

4.2.2.1.b Text Prompt Construction

Text prompts are created to describe the target organ or organ set for segmentation and serve as textual inputs to the model. To increase linguistic diversity, prompts are constructed with varying lengths and levels of complexity. For single-organ samples, common forms include:

- Short prompts containing only the organ name: "liver", "left kidney", "gallbladder".
- Medium-length instructional prompts: "segment the liver", "identify the pancreas".
- Longer anatomical sentences: "please identify and segment the urinary bladder in this CT scan".

For multi-organ samples, the prompts explicitly mention all target organs in a single expression so that the model learns to associate one textual query with multiple foreground regions. Representative forms include short prompts such as "liver and pancreas" or "left kidney and spleen", Medium-length instructional prompts such as "segment the liver and pancreas", and longer requests such as "please identify and segment the left kidney together with the spleen in this CT scan".

This design allows the textual branch to represent not only isolated anatomical targets but also composite anatomical queries.

To improve robustness to imperfect textual inputs, text noise is introduced by randomly modifying a small number of characters in organ names. Examples include "liver" → "livar" and "left kidney" → "left kidnez". These noisy organ names are incorporated into both single-organ and multi-organ prompt templates to simulate typographical errors and spelling variations.

By combining single-organ prompts, multi-organ prompts, and noisy textual variations, the model becomes more robust to diverse input styles, compound anatomical requests, and minor spelling mistakes, thereby improving its generalization ability in real-world applications.

These prompt texts are stored in text files and paired with the corresponding image and label files.

4.2.2.1.c Sample Creation Workflow

Each sample is assembled through the following steps:

1. Load a raw 3D CT volume and its full multi-organ segmentation label.
2. Select one or more target organs by filtering the label volume to include only the desired organ ID(s).
3. Create a text prompt describing the selected organ(s).
4. Save the processed image, the filtered binary mask, and the prompt text as a unified training sample.

4.2.2.1.d Organ-specific Sample Counts

The dataset contains both single-organ and multi-organ examples. The approximate number of samples per organ is:

- **180 samples**: Seminal_vesicles, Adrenal_Left, Adrenal_Right, Gallbladder, Esophagus, Duodenum, Rectum, Pancreas.
- **100 samples**: Prostate, Kidney_Left, Kidney_Right, Spleen, Bladder, Stomach, Colon, Intestine, Liver, Head_of_femur_Left, Head_of_femur_Right.
- **700 multi-organ samples**: Combinations such as "liver and pancreas" or "left kidney and spleen".

These counts reflect the dataset composition and help explain why the model must handle organs with very different frequencies and sizes.

4.2.2.1.e Dataset Directory Layout

The processed data is organized into train, validation, and test splits, with each organ stored in its own directory. The directory structure is:

```
data/
|-- train/
|   |-- Organ1/
|   |   |-- Image/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   |-- Label/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   \-- Text/
|   |       |-- case1.txt
|   |       |-- case2.txt
|   |       \-- ...
|   |-- Organ2/
|   |   |-- Image/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   |-- Label/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   \-- Text/
|   |       |-- case1.txt
|   |       |-- case2.txt
|   |       \-- ...
|   \-- ...
|-- val/
|   |-- Organ1/
|   |   |-- Image/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   |-- Label/
|   |   |   |-- case1.nii.gz
```

```
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   \-- Text/
|   |       |-- case1.txt
|   |       |-- case2.txt
|   |       \-- ...
|   |-- Organ2/
|   |   |-- Image/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   |-- Label/
|   |   |   |-- case1.nii.gz
|   |   |   |-- case2.nii.gz
|   |   |   \-- ...
|   |   \-- Text/
|   |       |-- case1.txt
|   |       |-- case2.txt
|   |       \-- ...
|   \-- ...
\-- test/
    |-- Organ1/
    |   |-- Image/
    |   |   |-- case1.nii.gz
    |   |   |-- case2.nii.gz
    |   |   \-- ...
    |   |-- Label/
    |   |   |-- case1.nii.gz
    |   |   |-- case2.nii.gz
    |   |   \-- ...
    |   \-- Text/
    |       |-- case1.txt
    |       |-- case2.txt
    |       \-- ...
    |-- Organ2/
    |   |-- Image/
    |   |   |-- case1.nii.gz
    |   |   |-- case2.nii.gz
    |   |   \-- ...
    |   |-- Label/
```

```
|   |   |-- case1.nii.gz
|   |   |-- case2.nii.gz
|   |   \-- ...
|   \-- Text/
|       |-- case1.txt
|       |-- case2.txt
|       \-- ...
\-- ...
```

This structure ensures that each organ's images, labels, and prompts remain aligned with minimal risk of mismatch.

4.2.2.2 Spatial Standardization and Resampling

To make examples comparable across different patients and scanners, all volumes are standardized in spatial resolution and orientation.

4.2.2.2.a Organ-specific Spacing

The preprocessing configuration was adjusted across different training stages to investigate the effect of voxel spacing on segmentation performance, particularly for small anatomical structures.

- **Stage 1**: A spacing of (1.5, 1.5, 1.5) mm was applied to all organs. This configuration reduced memory consumption and accelerated preprocessing and training, while still preserving sufficient anatomical information for medium and large organs.
- **Stage 2**: The isotropic spacing was reduced to (1.2, 1.2, 1.2) mm for all organs. The purpose of this adjustment was to preserve finer anatomical details and improve the representation of small organs. Although this configuration increased storage usage and computational cost, it provided higher spatial resolution and helped the model retain more boundary information during training.

This staged spacing strategy was used to analyze the trade-off between computational efficiency and segmentation quality across organs of different sizes.

4.2.2.2.b Resampling Transform

The preprocessing pipeline calls `Spacingd` with the selected `pixdim` value, applying:

- `bilinear` resampling for the image volumes.

- `nearest` neighbor resampling for the label masks.

This combination preserves soft tissue intensity continuity in the CT images while maintaining the discrete labels of the segmentation masks.

4.2.2.2.c Orientation Alignment

All image and label volumes are reoriented to the RAS coordinate system with `Orientationd`. This step eliminates discrepancy in axis ordering across datasets and ensures that the volume axes are consistent before training.

4.2.2.3 Intensity Pre-processing (Windowing)

The dataset applies an intensity window to reduce irrelevant CT ranges and focus on the soft tissue values relevant to organ segmentation.

4.2.2.3.a HU Clipping and Normalization

The raw CT image intensities are clipped from Hounsfield Units to the interval $[-175, 250]$. This windowing strategy removes extreme values from bone and air and concentrates the pixel intensities on the range where abdominal organs have the most meaningful contrast.

4.2.2.3.b Scale Intensity to [0, 1]

After clipping, the pipeline scales image intensities linearly to the range $[0.0, 1.0]$ using `ScaleIntensityRanged`. This normalization supports stable gradient descent and makes the input distribution consistent across the dataset.

4.2.2.4 Addressing Small Organ Imbalance

Many organs in the dataset are small and occupy only a small fraction of the volume. Without careful sampling, the model would see mostly background voxels during training and would struggle to learn the minority organ classes.

4.2.2.4.a Patch Sampling Strategy

During training, the data loader uses the MONAI transform `RandCropByPosNegLabeld` with:

- `spatial_size=(96, 96, 96)`: 3D patch size.
- `pos=3`: preference for positive patches containing the organ label.

- `neg=1`: still retains negative patches to preserve background context.
- `num_samples=4`: extracts four patches from each volume.

This sampling balances organ-containing patches with background patches, ensuring the network trains on both foreground anatomy and surrounding context.

4.2.2.4.b Motivation for Oversampling Small Organs

Small organs appear less frequently and occupy fewer voxels. Accordingly, the pipeline increases the probability that these organs appear in training patches, which improves the learning signal and reduces class imbalance.

4.2.2.5 Dataset Partitioning

The data is divided into training, validation, and test sets. Although exact split counts are determined by the processed dataset, the approximate distribution is:

Organ	Train	Val	Test
Seminal_vesicles	150	20	10
Adrenal_Left	150	20	10
Adrenal_Right	150	20	10
Gallbladder	150	20	10
Esophagus	150	20	10
Duodenum	150	20	10
Rectum	150	20	10
Pancreas	150	20	10
Prostate	70	20	10
Kidney_Left	70	20	10
Kidney_Right	70	20	10
Spleen	70	20	10
Bladder	70	20	10
Stomach	70	20	10
Colon	70	20	10
Intestine	70	20	10
Liver	70	20	10
Head_of_femur_Left	70	20	10
Head_of_femur_Right	70	20	10
Multi_organ	490	140	70

These splits reflect a practical balance between training volume and validation/testing coverage, with multi-organ examples providing additional variety for text-conditioned segmentation.

4.2.2.6 Processing Pipeline

The preprocessing pipeline is executed once before training and saves standardized volumes to a separate directory structure. This avoids repeated resampling and normalization during each epoch, which saves time and ensures consistency.

4.2.2.6.a Preprocessing Execution

For each organ folder, the pipeline:

1. Reads image and label file paths using `load_organ_data`.
2. Copies the existing text prompt files into the preprocessed `Text` folder.
3. Applies the organ-specific spatial transform and intensity normalization.
4. Saves the resulting preprocessed image and label volumes using `SaveImaged`.

4.2.2.6.b Dataset Consistency

The saved preprocessed data retains the same hierarchical structure as the raw dataset, with separate `Image`, `Label`, and `Text` folders under each organ. This design preserves compatibility with the training pipeline and simplifies dataset construction.

4.2.2.6.c Resource Efficiency

By saving preprocessed volumes before training, the pipeline reduces I/O and compute overhead during training. It also guarantees that every training and validation run uses identical spatially standardized and intensity-normalized inputs.

4.2.2.7 Summary

Our pipeline therefore accomplishes three essential goals:

- Creates paired multi-modal training samples with aligned text, image, and label data.
- Standardizes spatial resolution, orientation, and intensity to make volumes comparable across organs and patients.
- Addresses class imbalance through organ-aware sampling and careful split partitioning.

These preprocessing decisions form the foundation for the proposed multi-modal segmentation network.

4.3 Proposed Architecture

4.3.1 Overview of the Proposed Architecture

The model that my team proposed is a multi-modal medical image segmentation model, combining 3D CT image data and text prompts. The goal of the model is to leverage semantic information from text to guide the segmentation of specific anatomical organs.

As in the overall architecture, the model comprises three main components:

1. **Image Encoder (SwinUNETR)**: Extracts hierarchical image features from the input 3D CT block ($H \times W \times D$).
2. **Text Encoder (BioBERT)**: Encodes instructions text into embeddings carrying medical semantics.
3. **Fusion Module and Decoder**: At the bottleneck, image and text features are fused via the Fusion module. The Decoder then combines with skip connections from the Encoder to restore spatial resolution and provide a final segment prediction at the original size ($H \times W \times D$).

4.3.1.1 Model Architecture Implementation

The `MultiModalSegModel` class integrates all components into a cohesive PyTorch module:

4.3.1.1.1 Initialization Parameters

- `in_channels=1`: Accommodates single-channel CT images.
- `out_channels=1`: Produces binary segmentation masks.
- `feature_size=48`: Determines the channel progression in encoder/decoder.
- `embed_dim=768`: Unified embedding space for cross-modal fusion.
- `text_model_name`: Specifies the pre-trained biomedical language model.
- `num_heads=8`: Attention heads for fusion mechanism.



Figure 4.4: Architecture of the Multi-Modal Segmentation

4.3.1.1.2 Forward Pass Flow The forward method orchestrates the entire pipeline:

- Image encoding: `img_seq, spatial_shape, skips = image_encoder(image)`
- Text encoding: `txt_seq = text_encoder(text_inputs)`
- Cross-modal fusion: `fused_seq, _ = fusion(img_seq, txt_seq)`
- Feature reshaping: Converts sequence back to 3D volume.
- Decoding: `output = decoder(fused_vol, skips)`

4.3.1.2 Training and Inference Pipeline

4.3.1.2.1 Training Phase

- Data loading with MONAI DataLoader for efficient batching.
- Text tokenization using BioBERT tokenizer.
- Forward pass through the multi-modal model.
- Loss computation using DiceCELoss.
- Backpropagation and optimizer step.
- Validation with sliding window inference for full volumes.

4.3.1.2.2 Inference Phase

- Sliding window inference to handle large 3D volumes.
- Text prompt processing and tokenization.
- Model prediction with sigmoid activation.
- Post-processing: thresholding, connected component analysis.
- Inverse transforms to restore original image space.

4.3.1.3 Key Innovations

- **Multi-Modal Integration**: Combines vision and language modalities for guided segmentation.
- **Cross-Attention Fusion**: Dynamic attention mechanism for semantic-image alignment.
- **Pre-trained Components**: Leverages SwinUNETR and BioBERT for robust feature extraction.
- **Memory Efficient**: Uses sliding window inference and gradient checkpointing.
- **Flexible Text Prompts**: Supports various anatomical descriptions.

4.3.1.4 Technical Specifications

- **Input**: 3D CT volumes ($H \times W \times D \times 1$), text prompts.
- **Output**: Binary segmentation masks ($H \times W \times D \times 1$).
- **Framework**: PyTorch with MONAI and Transformers.
- **GPU Memory**: Optimized for sliding window processing.
- **Precision**: Mixed precision training support.

This model provides a comprehensive solution for text-guided medical image segmentation, enabling precise organ delineation through natural language instructions.

4.3.2 Image Encoder: SwinUNETR Architecture

The model's image input utilizes the SwinUNETR architecture, a Transformer-based neural network specifically designed for 3D medical imaging, which provides better global context capture than traditional CNNs.

4.3.2.1 Swin Transformer Block Implementation

Swin Transformer Block applies a self-attention mechanism that performs computations within non-overlapping local windows, rather than across the entire image block, significantly reducing computational complexity from second to first order.

This process comprises two consecutive modules: Window Multi-head Self-Attention ($W-MSA$) and Shifted Window Multi-head Self-Attention ($SW-MSA$). Shifting the windows allows for the connection of information between adjacent windows in subsequent transformation layers, ensuring the model learns continuous anatomical structures in 3D space.

4.3.2.2 3D Segmentation Adaptation for CT Volumes

The input 3D CT block of size ($H \times W \times D$) is divided into small patches and passed through Swin Transformer blocks. Feature extraction takes place in 4 hierarchical stages, gradually reducing spatial resolution and increasing the number of feature channels (channel dimensions), specifically:

- Stage 1: Resolution $\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2}$, number of channels: 48.
- Stage 2: Resolution $\frac{H}{4} \times \frac{W}{4} \times \frac{D}{4}$, number of channels: 96.
- Stage 3: Resolution $\frac{H}{8} \times \frac{W}{8} \times \frac{D}{8}$, number of channels: 192.
- Stage 4: Resolution $\frac{H}{16} \times \frac{W}{16} \times \frac{D}{16}$, number of channels: 384.

The features at each stage are not only propagated forward but also stored for skip connections to the decoder.

4.3.2.3 Implementation Details

In the implementation, the SwinUNETR is instantiated with the following key parameters:

- `in_channels=1`: Single-channel input for grayscale CT images.
- `out_channels=1`: Single-channel output for binary segmentation.
- `feature_size=48`: Base feature size, determining the channel progression.
- `depths=(2, 2, 2, 2)`: Number of Swin Transformer blocks per stage.
- `num_heads=(3, 6, 12, 24)`: Number of attention heads per stage, increasing with depth.
- `window_size=7`: Size of the local attention window.
- `dropout_path_rate=0.1`: Dropout rate for stochastic depth regularization.
- `use_checkpoint=True`: Memory-efficient checkpointing for large models.

The bottleneck features from Stage 4 are flattened and projected to a higher-dimensional embedding space using a linear layer that maps from 384 dimensions to 768 dimensions, aligning with the text encoder's output dimension for cross-modal fusion.

4.3.2.4 Feature Sequence Generation

The bottleneck feature map of shape $(B, 384, D', H', W')$ is reshaped into a sequence of tokens $(B, N, 384)$, where $N = D' \times H' \times W'$. This sequence is then projected to $(B, N, 768)$ using the linear projection layer. The spatial dimensions (D', H', W') are preserved for later reconstruction in the decoder.

Skip connections from all four stages are maintained, each containing hierarchical features that will be concatenated with corresponding decoder features to recover spatial details lost during downsampling.

4.3.3 Text Encoder: BioBERT Integration

To enable the model to "understand" which organ segment to segment, text prompts are processed through a large, specialized medical language model.

4.3.3.1 Pre-training and Fine-tuning Strategy

In this project, we use the dmis-lab/biobert-base-cased-v1.1 model, a variant of the BERT model pre-trained on the massive biomedical text dataset (PubMed). BioBERT encodes the input text string and extracts feature vectors at the token [CLS]. The output of the Text Encoder is a 1D vector of length 768 (768-dimensional embedding) containing high-density semantic information about the organ to be segmented. During training the segmentation model, the BioBERT weights are frozen or fine-tuned depending on the learning strategy.

4.3.3.2 Representing Anatomical Segmentation Instructions

Text instructions are represented as short text sentences, e.g., "liver", "left kidney", longer text sentences or combined phrases (multi-organ) if the image contains multiple organs to be segmented. The text input is processed by Tokenizer with a fixed maximum length (padding and truncation max_length = 48) before being fed into BioBERT to ensure uniformity in tensor size during batch training.

4.3.3.3 Implementation Details

The TextEncoder class initializes the BioBERT model using the Hugging Face transformers library:

- `model_name="dmis-lab/biobert-base-cased-v1.1"`: Pre-trained BioBERT model optimized for biomedical text.
- The model's `last_hidden_state` is extracted, which has shape $(B, T, 768)$, where B is batch size, T is sequence length, and 768 is the hidden dimension.
- A linear projection layer maps the 768-dimensional features to the target embedding dimension (also 768 in this implementation) for consistency with the image encoder's output.

4.3.3.4 Text Processing Pipeline

During training and inference:

- Text prompts are tokenized using the AutoTokenizer from the same BioBERT model.
- Tokenization includes padding to max_length=48 and truncation of longer sequences.
- Special tokens ([CLS], [SEP]) are added automatically by the tokenizer.
- The tokenized inputs are converted to PyTorch tensors and moved to the appropriate device.
- BioBERT processes the tokenized inputs to produce contextual embeddings.
- The [CLS] token embedding (first token) or the full sequence is used for downstream fusion.

4.3.3.5 Freezing Strategy

To prevent catastrophic forgetting and leverage pre-trained biomedical knowledge, the BioBERT parameters are frozen during initial training phases. This is implemented by setting `requires_grad = False` for all parameters in the `text_model`. In later fine-tuning stages, selective unfreezing may be applied to adapt the language model to the specific segmentation task.

4.3.4 Cross-Modal Fusion and Alignment

The model implements a dynamic fusion mechanism between 3D image features and linguistic features at the network bottleneck. Instead of using static operations, this study applies a Cross-Attention mechanism to allow images to actively "query" semantic information from text.

4.3.4.1 Image-Text Feature Space Mapping

Before implementing cross-attention, features from the Image Encoder and Text Encoder need to be brought into the same embedding space.

- Image Sequence Projection: Image features from Stage 4 are flattened into a sequence of tokens. Here, a linear projection layer is used to convert the number of channels from 384 to 768.

- Text Embedding: The output vector from BioBERT (1×768) is considered the target sequence containing anatomical information. Simulating the embedding size $d_model = 768$ for both methods is a prerequisite for implementing Multi-head Attention.

4.3.4.2 Cross-Modal Fusion via Multi-head Cross-Attention

The fusion mechanism is implemented through the CrossModalFusion module, using Multi-head Attention (MHA) structure to establish correlations between spatial regions in the image and keywords in the text directive.

A. Query-Key-Value (Q-K-V) mechanism:

In this architecture, the components of Attention are defined as follows:

- Query (Q): Extracted from the image feature string (img_seq). The image acts as the entity actively searching for relevant information.
- Key (K) and Value (V): Extracted from the embedded text (txt_seq). The text acts as a semantic repository providing segmentation guidance.

The Attention operation is performed using the following formula:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Here, the model uses 8 attention heads (num_heads=8) working in parallel, allowing it to focus on multiple different semantic aspects of the anatomical organ simultaneously.

B. Residual Connection and Layer Normalization:

To ensure stability during training and preserve original spatial features, the module employs a Residual Connection structure combined with Layer Normalization:

$$F_{\text{fused}} = \text{LayerNorm}\left(F_{\text{img}} + \text{MultiHeadAttention}(F_{\text{img}}, F_{\text{txt}}, F_{\text{txt}})\right)$$

The result is a feature block that has been "fused" with text semantics, in which each voxel now carries weighted information about its relevance to the input anatomical indicator.

C. Batch Expansion Logic:

This module is designed to flexibly handle the case where a single text instruction is applied to a batch of multiple images (Batch size ¿ 1). Through the expand operation, the text vector is copied proportionally to the batch size of the images, ensuring consistency in the neural network's matrix operations.

D. Interpretability:

In addition to generating fusion features, this mechanism also returns attention weights. This allows for visualization analysis to see which spatial region within the CT scan the model is actually "looking" at when receiving a specific text prompt, increasing transparency in medical diagnosis.

4.3.4.3 Implementation Details

The CrossModalFusion class is implemented as follows:

- `embed_dim=768`: Embedding dimension matching both image and text features.
- `num_heads=8`: Number of attention heads for multi-head attention.
- `batch_first=True`: Input tensors have batch dimension first.
- The attention mechanism uses PyTorch's `nn.MultiheadAttention` with default scaling and no dropout.

4.3.4.4 Batch Size Handling

The fusion module handles variable batch sizes dynamically:

- If the text sequence has a batch size of 1 while the image sequence has a larger batch size, the text sequence is duplicated along the batch dimension so that it matches the size of the image batch. This is done by expanding the text tensor to have the same first dimension as the image sequence.
- Otherwise, batch sizes must match exactly, raising a ValueError if mismatched.

This ensures flexibility in training scenarios where multiple images may share the same text prompt or require individual prompts.

4.3.5 Decoder

After passing through the CrossModalFusion module, the feature tensor F_{fused} is reshaped from a sequence back into a 3D volume with dimensions $\frac{H}{16} \times \frac{W}{16} \times \frac{D}{16}$ and 768 color channels. This semantically rich feature volume is then fed into the decoder to begin the decoding process and restore its spatial resolution.

4.3.5.1 Upsampling and Skip Connections

The decoder consists of multiple transposed convolution stages to progressively increase spatial resolution and reduce the number of channels. The decoding process is symmetrical with the encoder:

- The Fused Volume (768 channels) is upsampled and downscaled to 384 channels, then concatenates with the features from Stage 4 of the Encoder.
- This process is repeated at 192, 96, and 48 channels, with resolution increasing to $\frac{H}{8}$, $\frac{H}{4}$, $\frac{H}{2}$. The short circuits (blue arrows in the diagram) are crucial in restoring the lost organ boundary details during the downsample process.

4.3.5.2 Segmentation Head

In the final step, a convolutional layer $1 \times 1 \times 1$ is applied to the feature map with spatial resolution $\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2}$ (with 48 channels) and interpolated up to the original size to produce a binary or multi-layer prediction at size $(H \times W \times D)$. The output uses a Sigmoid activation function to determine the probability each voxel belongs to the authority specified by the text.

4.3.5.3 Implementation Details

The Decoder3D class implements as a U-Net style decoder with the following architecture:

- `in_channels=768`: Input channels from the fused features.
- `out_channels=1`: Single-channel binary segmentation output.
- `feature_size=48`: Base feature size for channel progression.

4.3.5.3.1 Bottleneck Processing The decoder begins with a bottleneck convolution that applies a linear transformation to map the 768 input channels to $48 \times 16 = 768$ channels, preparing for the upsampling stages.

4.3.5.3.2 UpSkipBlock3D Modules Each upsampling block consists of multiple classes responsible for doubling the spatial size, concatenating with corresponding skip connections from the encoder, and applying 3D convolutions with LeakyReLU activation and Instance Normalization.

The specific upsampling stages are:

1. `up3`: From 768 to 384 channels, concatenates with encoder skip3 (384 channels).
2. `up2`: From 384 to 192 channels, concatenates with encoder skip2 (192 channels).
3. `up1`: From 192 to 96 channels, concatenates with encoder skip1 (96 channels).
4. `up0`: From 96 to 48 channels, concatenates with encoder skip0 (48 channels).

4.3.5.3.3 Final Upsampling and Output Convolution After the skip connections, a final block increases resolution to $H \times W \times D$ while maintaining 48 channels.

Then, a $1 \times 1 \times 1$ convolution reduces to the final output channels (1 for binary segmentation).

4.3.5.4 Spatial Resolution Handling

The decoder handles potential spatial mismatches between upsampled features and skip connections using `nn.functional.interpolate` with trilinear mode and align_corners=False to ensure compatibility.

This architecture ensures that high-resolution spatial details from the encoder are effectively combined with the semantically enriched bottleneck features from the fusion module.

Chapter 5

EXPERIMENTAL SETUP AND RESULTS

5.1 Experimental Setup

5.1.1 Hardware and Software Environment

The model was developed using Python, employing the PyTorch Deep Learning framework in conjunction with the MONAI library, specifically designed for medical image segmentation. Training and testing were performed on high-performance computing systems, optimizing tensor operations with GPU support. Supporting libraries included Hugging Face transformers for language processing and nibabel for reading and writing standard biomedical data (.nii.gz).

5.1.2 Data Preprocessing Pipeline

The preprocessing pipeline transforms raw CT data into standardized formats suitable for training and inference.

5.1.2.1 Preprocessing Transforms

Raw data preprocessing uses MONAI transforms:

- `LoadImaged`: Loads .nii.gz files for images and labels.
- `EnsureChannelFirstd`: Reorders dimensions to (C, H, W, D).
- `Orientationd`: Reorients to RAS coordinate system.
- `Spacingd`: Resamples voxels using bilinear interpolation for images and nearest neighbor for labels.

- `ScaleIntensityRanged`: Normalizes HU values from [-175, 250] to [0, 1] range, with clipping.
- `SaveImaged`: Saves processed data in .nii.gz format.

5.1.2.2 Structure

Preprocessed data for each organ is organized as:

- `Image`: Processed CT volumes.
- `Label`: Ground truth segmentation masks.
- `Text`: Text prompts for each case.

5.1.2.3 Text Data Handling

Text prompts are copied directly from raw data directories and stored alongside corresponding images and labels, maintaining one-to-one correspondence for supervised learning.

5.1.3 Hyperparameters and Training Configuration

5.1.3.1 Model Architecture Parameters

The MultiModalSegModel is initialized with the following key parameters:

- `in_channels=1`: Single-channel input for CT images.
- `out_channels=1`: Single-channel binary segmentation output.
- `feature_size=48`: Base feature dimension for the SwinUNETR encoder.
- `embed_dim=768`: Embedding dimension for cross-modal fusion, matching BioBERT's hidden size.
- `text_model_name="dmis-lab/biobert-base-cased-v1.1"`: Pre-trained biomedical language model.
- `num_heads=8`: Number of attention heads in the fusion module.

5.1.3.2 Data Preprocessing and Augmentation

Data preprocessing is performed using MONAI transforms with the following configuration:

5.1.3.3 Training Transforms:

- `LoadImaged`: Loads image and label data from disk.
- `EnsureChannelFirstd`: Ensures channel dimension is first (C, H, W, D).
- `RandCropByPosNegLabeld`:
  - `keys`: Applies to both image and label.
  - `label_key`: Uses label for sampling guidance.
  - `spatial_size`: ROI size for patches.
  - `pos`: Ratio of positive samples (containing labels).
  - `neg`: Ratio of negative samples (background).
  - `num_samples`: Number of patches per input volume.
- `ToTensord`: Converts to PyTorch tensors.
- `EnsureTyped`: Ensures proper data types with metadata tracking.

5.1.3.4 Validation and Test Transforms:

- `LoadImaged`: Loads data without augmentation.
- `EnsureChannelFirstd`: Channel ordering.
- `ToTensord` and `EnsureTyped`: Tensor conversion.

5.1.3.5 Inference Transforms:

- `LoadImaged`: Loads test images.
- `EnsureChannelFirstd`: Channel ordering.
- `Orientationd`: Reorients to RAS coordinate system.
- `Spacingd`: Resamples voxels.
- `ScaleIntensityRanged`: Intensity scaling from [-175, 250] HU to [0, 1].
- `EnsureTyped`: Tensor preparation.

5.1.3.6 Optimization Configuration

- `Optimizer`: AdamW with `lr=1e-4` and `weight_decay=1e-5`.
- `Scheduler`: ReduceLROnPlateau with:
  - `mode='min'`: Monitors validation loss.
  - `factor=0.5`: Multiplies LR by 0.5 on plateau.
  - `patience=5`: Epochs to wait before reducing LR.
  - `threshold=1e-4`: Minimum change to qualify as improvement.
  - `min_lr=1e-6`: Minimum learning rate.

5.1.3.7 Training Loop Parameters

- `batch_size`: Training batch size.
- `num_workers`: DataLoader workers for training.
- `shuffle`: Shuffles training data.
- `pin_memory`: Pins memory for faster GPU transfer.
- `persistent_workers`: Keeps workers alive between epochs.
- `val_batch_size`: Validation batch size.
- `val_num_workers`: Validation DataLoader workers.

5.1.3.8 Text Processing Configuration

- `Tokenizer`: AutoTokenizer from BioBERT.
- `max_length=48`: Maximum sequence length with padding/truncation.
- `Tokenization`: Includes special tokens ([CLS], [SEP]).

5.1.3.9 Training Strategy

- `Early Stopping`: Patience of 8 epochs, minimum delta 1e-4.
- `Checkpointing`: Saves best model based on Dice score, latest model every epoch.
- `Resume Training`: Loads latest checkpoint if available.
- `Validation Interval`: Validates every epoch.
- `Additional Epochs`: The number of additional epochs to train on resume.

5.1.3.10 Evaluation Metrics

- `DiceMetric`: Binary Dice coefficient, background excluded.
- `HausdorffDistanceMetric`: 95th percentile HD95, background excluded.
- `Post-processing`: Activations(sigmoid=True), AsDiscrete(threshold=0.5).

5.1.3.11 Data Organization

- `Train/Val/Test Roots`: Separate directories for each split.
- `Organ-wise Structure`: Each organ has Image/, Label/, Text/ subdirectories.
- `Dataset Building`: Combines all organs into single dataset with shuffling.

5.1.4 Training Strategy and Monitoring

5.1.4.1 Encoder Freezing Strategy

To leverage pre-trained knowledge and prevent catastrophic forgetting:

- `freeze_encoders()`: Sets `requires_grad=False` for SwinUNETR and BioBERT parameters.
- Allows fusion and decoder parameters to be trained initially.
- Potential for fine-tuning encoders in later stages.

5.1.4.2 Checkpoint Management

- `best_checkpoint.pth`: Saved when validation Dice improves.
- `latest_checkpoint.pth`: Saved every epoch for resumption.
- Stores model, optimizer, scheduler states, and training metrics.
- Resume logic: Loads latest checkpoint.

5.1.4.3 Early Stopping Mechanism

- Monitors validation Dice score.
- `patience=8`: Epochs without improvement before stopping.
- `min_delta=1e-4`: Minimum improvement threshold.
- Prevents overfitting and saves computational resources.

5.1.4.4 Logging and History Tracking

- CSV history file with columns: run_id, epoch, train_loss, val_loss, val_dice, lr.
- Timestamped run IDs for experiment tracking.
- Automatic history file initialization.
- Learning rate logging for hyperparameter analysis.

5.1.4.5 Validation Strategy

- Performed every epoch with (`VAL_INTERVAL=1`).
- Uses full volumes without cropping.
- Sliding window inference for memory efficiency.
- Comprehensive metrics: Dice, loss.

5.2 Evaluation Metrics

To quantitatively assess the model's accuracy, the study used two primary measures in medical image processing, combined with appropriate loss functions to optimize the learning process.

5.2.1 Segmentation Metrics

- **Dice Similarity Coefficient (DSC)**: Evaluate the degree of overlap between the model's prediction and the original label (Ground Truth). The value ranges from 0 to 1, with a value closer to 1 being better.

$$DSC = \frac{2|A \cap B|}{|A| + |B|}$$

  In this case, A is the set of voxels that made correct predictions and B is the original label.

- **Hausdorff Distance 95th Percentile (HD95)**: Calculates the maximum distance between the surface of the predicted segment and the actual surface, after removing 5% of the confounding points (outliers). This measure is particularly important for assessing the accuracy of tumor or organ boundaries. A smaller value is better.

5.2.2 Training Loss Functions

Due to the significant imbalance between the number of voxels in the basal region (accounting for >95%) and the organ region, the choice of loss function is crucial:

- **DiceCELoss**: A combination of Dice Loss and Cross-Entropy Loss. Dice Loss optimizes overall overlap (directly maximizing the Dice Score), while Cross-Entropy penalizes pixel-level mispredictions. This function is effective for large-sized organs.
- **DiceFocalLoss**: A solution applied to small organs (such as Esophagus, gallbladder, ...). By combining Dice Loss and Focal Loss (with the γ parameter usually set to 2.0), this function forces the model to focus on hard samples at the boundaries, while minimizing noise caused by excessive background noise.

5.2.3 Implementation Details of Metrics

5.2.3.1 Dice Metric Configuration

- `include_background=False`: Excludes background class from calculation.
- `reduction="mean"`: Averages across all classes/samples.
- Computed on post-processed predictions (sigmoid + threshold).

5.2.3.2 Hausdorff Distance Configuration

- `include_background=False`: Excludes background.
- `percentile=95`: Uses 95th percentile to reduce outlier sensitivity.
- `reduction="mean"`: Averages across samples.
- Only computed when both prediction and ground truth have non-zero voxels.

5.2.3.3 Loss Function Implementation

The primary loss function used is MONAI's `DiceCELoss`:

- `sigmoid`: Applies sigmoid to predictions.
- `squared_pred`: Uses linear Dice computation.
- `batch`: Computes batch-wise loss.

- `lambda_dice`: Dice loss weight.
- `lambda_ce`: Cross-entropy loss weight.

In the second stage, for small organs, `DiceFocalLoss` is used with:

- `sigmoid`: Applies sigmoid activation.
- `gamma`: Focusing parameter for Focal Loss.
- `lambda_dice`: Dice loss weight.
- `lambda_focal`: Focal loss weight.
- `include_background`: Excludes background from loss calculation.

5.3 Result and Discussion

5.3.1 Quantitative Performance Analysis

This section examines the quantitative behavior of the proposed SwinUNETR–BioBERT framework on the RAOS dataset [**?**]. Within the conceptual scope of this thesis, the caption-like anatomical description can be interpreted as a semantic retrieval cue that interacts with volumetric image features through multimodal alignment. Rather than generating a ranked list of external images, the model identifies the anatomically relevant region inside a 3D volume and expresses that retrieval-oriented localization behavior as a binary mask. Under this interpretation, the reported metrics describe how effectively the semantic query activates the intended anatomical target and how spatially coherent the resulting localization remains.

All results in this chapter were obtained using sliding window inference with an ROI size of $96 \times 96 \times 96$, followed by sigmoid activation, thresholding at 0.5, and connected component filtering. The reported performance therefore reflects caption-conditioned anatomical localization after the same post-processing pipeline used during inference.

5.3.1.1 Evaluation Metrics

Three complementary metrics were used to quantify localization quality: Dice Similarity Coefficient (DSC), Normalized Surface Distance (NSD), and Hausdorff Distance at the 95th percentile (HD95).

Dice Similarity Coefficient (DSC ↑) measures the volumetric overlap between the predicted mask P and the ground-truth mask G:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{5.1}$$

A DSC value close to 100% indicates strong overlap, whereas a value close to 0% indicates that the predicted foreground does not meaningfully recover the target structure.

Normalized Surface Distance (NSD ↑) evaluates the fraction of the predicted surface that lies within a specified tolerance distance τ of the reference surface. NSD captures boundary accuracy more directly than DSC, as it focuses on surface proximity rather than volumetric overlap. NSD was not used as a training objective or validation loss in our framework; it is reported here solely for fair comparison with the RAOS benchmark results.

Hausdorff Distance at the 95th percentile (HD95 ↓) measures boundary discrepancy by evaluating the 95th percentile of the symmetric surface distances between the predicted surface S_P and the reference surface S_G:

$$\mathrm{HD95}(P, G) = \max\left\{Q_{0.95}\big(d(S_P, S_G)\big),\ Q_{0.95}\big(d(S_G, S_P)\big)\right\}, \tag{5.2}$$

where $Q_{0.95}(\cdot)$ denotes the 95th percentile of the corresponding surface-distance distribution. Lower HD95 values indicate better spatial compactness and more stable anatomical localization.

DSC, NSD, and HD95 are therefore complementary rather than interchangeable. DSC is primarily sensitive to volumetric overlap, NSD focuses on surface boundary quality, whereas HD95 is sensitive to spatial outliers and boundary irregularities. This distinction is important for retrieval-oriented localization: a prediction may recover more of the intended target region and thus improve DSC, yet still exhibit poor HD95 if small disconnected components appear far from the true anatomy. In such cases, semantic target recovery is present, but the retrieved region is not spatially compact.

5.3.1.2 Global Performance Comparison

Table 5.1 summarizes the global comparison between Stage 1 and Stage 2. Stage 1, evaluated at epoch 126, produced an overall mean DSC of 57.85% and an overall mean HD95 of 22.18. After the second-stage optimization, the mean DSC increased to 60.16% at epoch 176, corresponding to an absolute gain of 2.31 percentage points. The mean NSD, however, decreased slightly from 54.25% to 53.95%, and the mean HD95 increased from 22.18 to 42.39. This pattern indicates that Stage 2 improved volumetric overlap for several hard targets but at the cost of spatial compactness.

Table 5.1: Overall stage-wise comparison. DSC and NSD are reported in percentage (%). Stage 2 improved mean DSC but increased HD95, reflecting a trade-off between semantic sensitivity and spatial compactness.

Stage	Epoch	Loss	Pixdim	DSC ↑	NSD ↑	HD95 ↓	Key Observation
Stage 1	126	`DiceCE`	$1.5 \times 1.5 \times 1.5$	57.85	54.25	22.18	Stable large-organ localization; weak small-organ retrieval.
Stage 2	176	`DiceFocal`	$1.2 \times 1.2 \times 1.2$	60.16	53.95	42.39	Improved small-organ activation; fragmented masks increased HD95.

The global DSC difference between the two stages is modest (+2.31 pp), but the HD95 shift is substantial (+20.21). This divergence indicates that Stage 2 did not simply improve or degrade performance in a uniform manner. Instead, the optimization transition suggests a trade-off: the later configuration increased sensitivity to semantically specified targets, yet the resulting localization became less spatially compact and less boundary-stable. The caption-like anatomical description appears to support stronger query-conditioned activation, but semantic activation alone is not sufficient to guarantee compact retrieval-oriented localization.

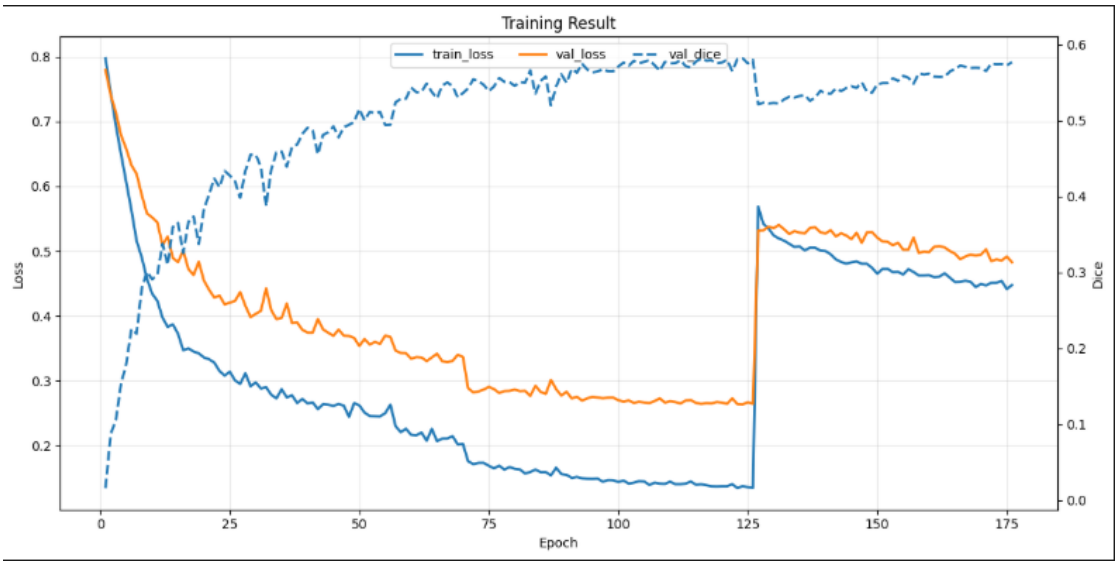


Figure 5.1: Stage 1 and Stage 2 training results.

5.3.1.3 Organ-wise Performance Trends

To make the optimization shift more interpretable, the organ-wise results were grouped into large, medium, and small organs. The DSC tables are sorted by

signed DSC change (Stage 2 − Stage 1) in descending order so that the strongest recoveries appear first. The HD95 tables are sorted by HD95 increase in descending order so that the most severe compactness failures appear first.

DSC-based overlap trends. The grouped DSC tables show where optimization emphasis shifted. Large organs remained the highest-overlap category in both stages, which indicates that broad anatomical regions were easier to recover from the caption-like semantic query. However, most large organs decreased in DSC after Stage 2. Instead, the largest overlap gains were concentrated in the small-organ group, particularly in the esophagus (+37.93 pp), gallbladder (+32.11 pp), and seminal vesicles (+13.15 pp). This pattern suggests that the second optimization stage redirected model capacity away from already stable dominant structures and toward semantically difficult, underrepresented targets.

Table 5.2: Large-organ DSC comparison (%) illustrating that Stage 2 redirected optimization emphasis away from already stable large-organ localization.

Organ	**Stage 1**	**Stage 2**	**Abs. Chg.**	**Rel. Chg.**	**Interpretation**
Stomach	74.53	75.53	+1.00	+1.3%	Mild overlap improvement
Right Kidney	85.74	84.55	−1.19	−1.4%	Stable overlap
Liver	93.30	91.47	−1.83	−2.0%	Stable compact localization
Left Femur Head	89.39	88.18	−1.21	−1.4%	Compact localization slightly weakened
Bladder	71.13	69.16	−1.97	−2.8%	Query-conditioned overlap reduced
Spleen	89.10	83.10	−6.00	−6.7%	Moderate overlap reduction
Left Kidney	84.57	76.53	−8.04	−9.5%	Overlap notably reduced
Right Femur Head	82.84	66.81	−16.03	−19.4%	Significant overlap degradation

Table 5.3: Medium-organ DSC comparison (%) showing that Stage 2 was not uniformly beneficial, with mostly stable overlap changes.

Organ	Stage 1	Stage 2	Abs. Chg.	Rel. Chg.	Interpretation
Prostate	67.02	68.29	+1.27	+1.9%	Mild overlap improvement
Colon	58.21	59.22	+1.01	+1.7%	Marginal improvement
Duodenum	45.81	46.51	+0.70	+1.5%	Marginal improvement
Pancreas	65.27	63.12	−2.15	−3.3%	Mild overlap degradation
Rectum	71.86	69.66	−2.20	−3.1%	Mild overlap degradation
Intestine	73.75	67.98	−5.77	−7.8%	Moderate overlap reduction

Table 5.4: Small-organ DSC comparison (%) demonstrating that the principal overlap gains of Stage 2 occurred in semantically difficult targets.

Organ	Stage 1	Stage 2	Abs. Chg.	Rel. Chg.	Interpretation
Esophagus	1.33	39.26	+37.93	+2851.9%	Strong semantic recovery
Gallbladder	8.14	40.25	+32.11	+394.5%	Strong semantic recovery
Seminal Vesicles	17.68	30.83	+13.15	+74.4%	Query-conditioned activation improved
Left Adrenal	0.00	5.61	+5.61	N/A	Retrieval sensitivity increased
Right Adrenal	0.00	3.29	+3.29	N/A	Semantic target recovery remained limited

The DSC tables therefore make the optimization transition visually explicit. Stage 1 preserved strong overlap on dominant anatomical regions, whereas Stage 2 shifted the most meaningful overlap gains toward small and semantically difficult structures. However, overlap alone does not reveal whether the retrieved regions remained compact. For this reason, the grouped HD95 analysis is necessary to understand how the optimization transition changed boundary stability and fragmentation behavior.

HD95-based compactness trends. The grouped HD95 tables show that the compactness penalty of Stage 2 was not confined to small organs. Large organs with relatively strong DSC values still experienced severe HD95 growth, while several medium organs showed dramatic compactness collapse. This behavior suggests that the second optimization stage did not simply improve semantic target recovery at no cost; rather, it broadened foreground sensitivity in a way that also increased spatial outliers across multiple organ scales.

Table 5.5: Large-organ HD95 comparison showing that Stage 2 preserved some overlap while substantially weakening spatial compactness across large structures (**Lower is better**).

Organ	**Stage 1**	**Stage 2**	**Abs. Chg.**	**Rel. Chg.**	**Interpretation**
Spleen	20.73	188.39	+167.66	+808.8%	Severe fragmentation increase
Liver	6.05	46.05	+40.00	+661.2%	Compactness reduced despite high overlap
Right Femur Head	7.01	39.36	+32.35	+461.5%	Boundary stability reduced
Stomach	16.96	22.00	+5.04	+29.7%	Mild compactness degradation
Left Kidney	8.90	18.38	+9.48	+106.5%	Boundary stability reduced
Left Femur Head	4.04	9.36	+5.32	+131.7%	Compactness mildly degraded
Bladder	10.91	12.47	+1.56	+14.3%	Stable compactness
Right Kidney	10.05	8.81	−1.24	−12.3%	Compactness improved

Table 5.6: Medium-organ HD95 comparison illustrating that the Stage 2 sensitivity shift destabilized compact localization beyond the small-organ group (**Lower is better**).

Organ	**Stage 1**	**Stage 2**	**Abs. Chg.**	**Rel. Chg.**	**Interpretation**
Colon	59.22	78.90	+19.68	+33.2%	Moderate compactness degradation
Duodenum	21.12	33.84	+12.72	+60.2%	Boundary stability reduced
Intestine	15.98	27.61	+11.63	+72.8%	Mild compactness degradation
Pancreas	11.42	21.18	+9.76	+85.5%	Boundary stability reduced
Prostate	6.40	9.56	+3.16	+49.4%	Mild compactness degradation
Rectum	9.73	12.36	+2.63	+27.0%	Stable compactness

Table 5.7: Small-organ HD95 comparison showing that semantic recovery in Stage 2 was frequently accompanied by fragmentation and unstable compactness (**Lower is better**).

Organ	**Stage 1**	**Stage 2**	**Abs. Chg.**	**Rel. Chg.**	**Interpretation**
Seminal Vesicles	20.27	83.10	+62.83	+310.0%	Severe fragmentation increase
Left Adrenal	132.64	167.48	+34.84	+26.3%	Small-target retrieval remained unstable
Right Adrenal	128.59	155.82	+27.23	+21.2%	Small-target retrieval remained unstable
Esophagus	92.78	49.94	−42.84	−46.2%	Recovery accompanied by improved compactness
Gallbladder	75.03	45.44	−29.59	−39.4%	Recovery with improved boundary stability

The HD95 analysis sharpens the central narrative of this chapter. Stage 2 improved overlap for several small semantic targets, but the grouped compactness tables show that this sensitivity gain was accompanied by fragmentation in some organs while others (esophagus, gallbladder) actually improved in both DSC and HD95. The optimization transition is therefore best interpreted as a regime change: Stage 1 favored compact and conservative localization, whereas Stage 2 favored broader query-conditioned activation with variable effects on boundary stability depending on the organ.

5.3.2 Stage-wise Optimization Analysis

The two training stages can be interpreted as successive optimization regimes of retrieval-oriented localization behavior. Because the textual input serves as a caption-like anatomical description, the stage-wise comparison does not merely reflect a change in segmentation loss; it also reflects a change in how strongly the model responds to a semantic query and how compactly it localizes the retrieved anatomical region. From this perspective, Stage 1 emphasizes conservative and spatially stable localization, whereas Stage 2 emphasizes increased sensitivity to small, semantically difficult targets.

5.3.2.1 Stage 1: Initial Semantic Localization Behavior

Stage 1 corresponds to the initial optimization phase and was evaluated at epoch 126. The training configuration used a preprocessing pixdim of approximately $1.5 \times 1.5 \times 1.5$. Optimization employed MONAI's `DiceCELoss` with sigmoid activation, non-squared predictions, batch reduction, $\lambda_{\text{dice}} = 0.7$, and $\lambda_{\text{ce}} = 0.3$. Under this setting, the model achieved an overall mean DSC of 57.85% and an overall mean HD95 of 22.18.

The Stage 1 results suggest that the caption-like semantic retrieval cue was already sufficient to support stable localization for anatomically large structures. High DSC values were observed for the liver (93.30%), spleen (89.10%), left femur head (89.39%), and kidneys (84–86%), which indicates that broad regions with substantial voxel support were easier to recover through multimodal semantic alignment. Equally important, the comparatively low HD95 values (e.g., liver: 6.05, left femur head: 4.04) suggest that these localized regions remained spatially compact.

The main limitation of Stage 1 appears in the small-organ results. Both adrenal glands obtained DSC of 0.00%, while the esophagus achieved only 1.33% and the gallbladder 8.14%. These outcomes suggest that the semantic retrieval cue remained insufficient for several very small anatomical targets. In such cases, the model did not merely produce coarse boundaries; rather, it often failed to activate a meaningful foreground region at all. This interpretation is consistent with severe voxel imbalance, since tiny organs contribute a much smaller optimization signal than large structures.

The Stage 1 configuration may help explain this behavior. `DiceCELoss` combines overlap optimization with cross-entropy, which is often effective for dominant foreground regions but may provide less emphasis to rare, hard-to-detect structures. Likewise, the coarser resampling scale of $1.5 \times 1.5 \times 1.5$ may suppress fine anatomical detail that is important for semantic target recovery in very small organs. Overall, Stage 1 prioritized compact and stable localization behavior, but

remained insufficient for several semantically difficult small targets.

5.3.2.2 Stage 2: Small-Organ Sensitivity Optimization

Stage 2 extended optimization to epoch 176 and was designed specifically to improve the recovery of small anatomical targets. Three changes were introduced: the amount of small-organ training data was increased by approximately two-fold, the loss function was replaced with `DiceFocalLoss`, and the preprocessing pixdim was refined from $1.5 \times 1.5 \times 1.5$ to $1.2 \times 1.2 \times 1.2$. The `DiceFocalLoss` setting used sigmoid activation, $\gamma = 2.0$, $\lambda_{\text{dice}} = 1.0$, $\lambda_{\text{focal}} = 1.0$, and background exclusion. Collectively, these modifications were intended to increase exposure to underrepresented targets, preserve finer structural detail, and place greater optimization emphasis on difficult foreground regions.

The resulting DSC changes suggest that Stage 2 did strengthen sensitivity to several semantically specified small targets. The esophagus improved from 1.33% to 39.26%, the gallbladder from 8.14% to 40.25%, and the seminal vesicles from 17.68% to 30.83%. Both adrenal glands recovered from zero DSC to non-zero values (right: 3.29%, left: 5.61%). This pattern is consistent with stronger semantic target activation: once the optimization was biased toward rare and difficult structures, the caption-like anatomical description appears to provide a more effective retrieval-oriented signal for small target recovery.

At the same time, Stage 2 reduced DSC for many large and medium organs. The liver, spleen, kidneys, femur heads, bladder, and several other organs all declined to some extent. The right femur head experienced the most substantial decline (from 82.84% to 66.81%). This shift suggests that the later optimization stage did not uniformly strengthen image-text alignment across the entire organ set. Instead, it reallocated model sensitivity. Such behavior is consistent with the focal component of the loss, which places greater emphasis on hard examples and may therefore reduce the model's previous preference for smooth, dominant foreground regions.

Notably, several small organs improved in both DSC and HD95 simultaneously. The esophagus improved from HD95 of 92.78 to 49.94, and the gallbladder from 75.03 to 45.44, indicating that Stage 2 produced not only stronger overlap but also more spatially compact predictions for these targets. However, many other organs showed HD95 degradation, particularly the spleen (20.73 to 188.39) and seminal vesicles (20.27 to 83.10).

Stage 2 improved semantic sensitivity for difficult anatomical targets, but the effects on spatial compactness were organ-dependent. Some small organs improved in both DSC and HD95, while several large organs experienced fragmentation and boundary instability.

5.3.3 Per-Organ Analysis

To further analyze the segmentation behavior of the proposed text-guided framework across the two optimization stages, the organs were divided into three groups based on anatomical size: large organs, medium organs, and small organs. For each group, the comparison between Stage 1 and Stage 2 is presented using DSC (↑), NSD (↑), and HD95 (↓) to capture both volumetric overlap, surface boundary quality, and spatial compactness.

5.3.3.1 Large Organ Segmentation

Large organs include the liver, spleen, kidneys, stomach, bladder, and femur heads. These organs generally achieved the highest segmentation accuracy because of their larger spatial structures and clearer boundaries in CT images.

In Stage 1, the liver achieved the highest DSC of 93.30% with NSD of 85.13% and HD95 of 6.05, indicating both strong overlap accuracy and stable boundary localization. The spleen also performed well with DSC of 89.10% and NSD of 91.63%, although its HD95 of 20.73 suggests that boundary errors were still present in several difficult cases. The left and right kidneys achieved DSC of 84.57% and 85.74%, respectively, with comparable NSD values around 80–82%. The left femur head reached DSC of 89.39% with an excellent HD95 of 4.04, making it one of the most stable large organs.

In Stage 2, the liver maintained the best overlap among large organs with DSC of 91.47%, though NSD decreased to 75.05% and HD95 increased to 46.05. The right kidney showed slight improvement in HD95 (from 10.05 to 8.81), while most other large organs experienced increased HD95 values, particularly the spleen (from 20.73 to 188.39) and bladder (from 10.91 to 12.47). The stomach showed a marginal DSC increase from 74.53% to 75.53%, suggesting that Stage 2 refinement had a small positive effect on this organ's overlap.

Table 5.8: Performance of large organs across two training stages. DSC and NSD are reported in percentage (%). **Bold** indicates the better value between stages for each metric.

Organ	DSC (%) ↑		NSD (%) ↑		HD95 ↓	
	Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2
Liver	**93.30**	91.47	**85.13**	75.05	**6.05**	46.05
Spleen	**89.10**	83.10	**91.63**	87.06	**20.73**	188.39
Left Kidney	**84.57**	76.53	**80.83**	71.81	**8.90**	18.38
Right Kidney	**85.74**	84.55	**82.07**	78.60	10.05	**8.81**
Stomach	74.53	**75.53**	**62.73**	57.51	**16.96**	22.00
Bladder	**71.13**	69.16	**54.41**	48.06	**10.91**	12.47
Left Femur Head	**89.39**	88.18	**87.22**	82.21	**4.04**	9.36
Right Femur Head	**82.84**	66.81	**75.33**	59.47	**7.01**	39.36

5.3.3.2 Medium Organ Segmentation

Medium organs include the pancreas, colon, rectum, prostate, intestine, and duodenum. Compared with large organs, these structures are more difficult to segment because of irregular shapes and less distinct boundaries.

In Stage 1, the intestine achieved the highest DSC of 73.75% with NSD of 73.37% and HD95 of 15.98, indicating relatively reliable overlap and moderate boundary accuracy. The rectum followed with DSC of 71.86% and a notably low HD95 of 9.73, suggesting compact predictions despite moderate overlap. The prostate achieved DSC of 67.02% with NSD of 62.89% and an excellent HD95 of 6.40, making it one of the most spatially compact medium-organ predictions.

In Stage 2, the prostate showed mild DSC improvement (from 67.02% to 68.29%), while the duodenum also improved marginally (from 45.81% to 46.51%). The colon maintained its DSC at around 59% but its HD95 increased from 59.22 to 78.90, indicating reduced boundary compactness. The intestine experienced a decline in DSC from 73.75% to 67.98%, suggesting that the Stage 2 optimization shift toward small organs partially reduced medium-organ performance.

Table 5.9: Performance of medium organs across two training stages. DSC and NSD are reported in percentage (%). **Bold** indicates the better value between stages for each metric.

Organ	DSC (%) ↑		NSD (%) ↑		HD95 ↓	
	Stage 1	**Stage 2**	**Stage 1**	**Stage 2**	**Stage 1**	**Stage 2**
Pancreas	**65.27**	63.12	**62.92**	57.72	**11.42**	21.18
Colon	58.21	**59.22**	**52.38**	48.91	**59.22**	78.90
Intestine	**73.75**	67.98	**73.37**	63.11	**15.98**	27.61
Duodenum	45.81	**46.51**	**47.16**	43.36	**21.12**	33.84
Rectum	**71.86**	69.66	**66.46**	58.50	**9.73**	12.36
Prostate	67.02	**68.29**	**62.89**	54.75	**6.40**	9.56

5.3.3.3 Small Organ Segmentation

Small organs include the adrenal glands, gallbladder, esophagus, and seminal vesicles. These organs are considered the most challenging structures because of their small size and unclear boundaries.

In Stage 1, both adrenal glands obtained DSC and NSD of 0.00%, showing that the model could not identify these structures at the initial resolution setting. The esophagus achieved a DSC of only 1.33% with NSD of 2.75%, indicating almost no meaningful overlap. The gallbladder obtained DSC of 8.14% and NSD of 10.15%, which, although marginally better than the adrenal glands, still reflects highly unreliable segmentation. The seminal vesicles achieved DSC of 17.68% and NSD of 20.84% with HD95 of 20.27.

In Stage 2, several small organs showed substantial improvement. The esophagus improved dramatically from DSC of 1.33% to 39.26%, with NSD increasing from 2.75% to 48.71% and HD95 improving from 92.78 to 49.94. The gallbladder also improved from 8.14% to 40.25% DSC. The seminal vesicles reached DSC of 30.83%, though HD95 increased from 20.27 to 83.10, indicating spatial fragmentation. The adrenal glands showed marginal recovery (right: 3.29% DSC, left: 5.61% DSC) but remained extremely challenging, with HD95 values exceeding 150.

Table 5.10: Performance of small organs across two training stages. DSC and NSD are reported in percentage (%). **Bold** indicates the better value between stages for each metric.

Organ	**DSC (%) ↑**		**NSD (%) ↑**		**HD95 ↓**	
	Stage 1	**Stage 2**	**Stage 1**	**Stage 2**	**Stage 1**	**Stage 2**
Right Adrenal	0.00	**3.29**	0.00	**4.78**	128.59	155.82
Left Adrenal	0.00	**5.61**	0.00	**4.94**	132.64	167.48
Esophagus	1.33	**39.26**	2.75	**48.71**	92.78	**49.94**
Gallbladder	8.14	**40.25**	10.15	**43.45**	75.03	**45.44**
Seminal Vesicles	17.68	**30.83**	20.84	**30.83**	**20.27**	83.10

5.3.4 Comparison with State-of-the-Art Methods

To position the proposed text-guided framework within the broader landscape of abdominal organ segmentation, this section compares it against seven state-of-the-art methods evaluated on the RAOS benchmark [**?**]. The compared methods span a variety of architectural paradigms: nnU-Net [**?**] (self-configuring baseline SOTA), RepUX-Net [**?**] (CNN-based), 3D UX-Net [**?**] (ConvNeXt-based), nnFormer [**?**] (pure volumetric Transformer), SwinUNETR [4] (hierarchical Transformer), TransBTS [**?**] (hybrid CNN–Transformer), and UNETR [**?**] (pure Transformer encoder with CNN decoder). All SOTA results are taken directly from the RAOS benchmark paper.

An important distinction must be emphasized: all seven reference methods operate under a **fully supervised, class-index-based paradigm** in which each organ is identified by a fixed integer label during training and inference. In contrast, our proposed framework performs **text-guided anatomical retrieval**, where a natural-language prompt such as *"Identify the Liver using spatial features"* serves as the query to localize the target organ. This means our model does not require a predefined class index; instead, it can retrieve any organ specified by a textual description, offering greater flexibility and extensibility to unseen targets.

Two evaluation metrics are reported for all methods: Dice Similarity Coefficient (DSC) and Normalized Surface Distance (NSD). DSC measures volumetric overlap between the predicted and ground-truth masks. NSD evaluates the fraction of the predicted surface that lies within a tolerance distance of the reference surface, providing a complementary view of boundary accuracy. For our proposed method, Hausdorff Distance at the 95th percentile (HD95) is additionally reported to provide deeper insight into spatial compactness and boundary outlier behavior. Note that NSD was not used as a training objective or validation criterion in our framework; it is reported here solely for fair comparison with the SOTA benchmark results.

In the following table, **bold** indicates the best score, underline indicates the second best, and *italic* indicates the third best across all methods. These rankings are preserved from the original RAOS benchmark evaluation.

5.3.4.1 Overall Analysis

Table 5.11 reveals a substantial performance gap between the proposed text-guided framework and the fully supervised SOTA methods. The best-performing SOTA method, nnU-Net [**?**], achieves a mean DSC of 86.65% and mean NSD of 86.87%, whereas our best configuration (Stage 2) reaches a mean DSC of 60.16% and mean NSD of 53.95%. This gap of approximately 26 percentage points in DSC is significant and must be interpreted in the proper context.

Table 5.11: Comprehensive comparison of organ segmentation methods on the RAOS test set across all 19 organs. SOTA results are reported from the RAOS benchmark. Our method performs text-guided anatomical retrieval rather than class-index-based segmentation.

Organ	nnU-Net [?]		RepUX-Net [?]		3D UX-Net [?]		nnFormer [?]		SwinUNETR [4]		TransBTS [?]		UNETR [?]		Ours (Stage 1) (1.5 mm Resolution)			Ours (Stage 2) (1.2 mm Refined)		
	DSC↑	NSD↑	DSC↑	NSD↑	DSC↑	NSD↑	DSC↑	NSD↑	DSC↑	NSD↑	DSC↑	NSD↑	DSC↑	NSD↑	DSC↑	NSD↑	HD95↓	DSC↑	NSD↑	HD95↓
Liv	**96.65**	**90.59**	96.51	90.04	*96.43*	89.65	95.95	87.43	96.40	*89.69*	96.03	87.48	95.95	87.11	93.30	85.13	6.05	91.47	75.05	46.05
Spl	**95.93**	**96.91**	95.85	96.73	95.58	96.17	94.69	93.93	*95.72*	*96.44*	95.23	95.31	94.67	93.09	89.10	91.63	20.73	83.10	87.06	188.39
LKid	**95.80**	**96.44**	95.59	95.13	95.51	95.60	94.57	93.62	*95.56*	95.77	95.18	95.08	94.73	93.47	84.57	80.83	8.90	76.53	71.81	18.38
RKid	**95.56**	**96.13**	95.21	94.61	95.52	96.00	94.53	93.56	*95.35*	*95.49*	94.87	93.77	94.62	93.52	85.74	82.07	10.05	84.55	78.60	8.81
Sto	**92.77**	**86.99**	91.92	*84.09*	92.17	84.58	90.78	79.73	*92.04*	83.97	91.71	82.71	90.21	78.77	74.53	62.73	16.96	75.53	57.51	22.00
Gal	**80.63**	**83.87**	76.02	75.04	77.15	78.78	71.07	69.35	75.76	*76.12*	72.24	72.13	68.11	67.03	8.14	10.15	75.03	40.25	43.45	45.44
Eso	**83.23**	**90.78**	78.96	85.70	*78.57*	86.08	71.60	79.24	78.18	*85.89*	75.53	83.16	70.69	77.31	1.33	2.75	92.78	39.26	48.71	49.94
Pan	**83.28**	**82.89**	81.82	80.75	81.62	*80.65*	77.16	73.09	*81.74*	80.39	78.80	75.55	77.88	73.43	65.27	62.92	11.42	63.12	57.72	21.18
Duo	**72.05**	**73.00**	68.25	*68.57*	*68.42*	68.38	64.12	61.69	68.72	68.75	65.31	65.00	63.26	59.14	45.81	47.16	21.12	46.51	43.36	33.84
Col	**87.10**	**84.74**	85.40	80.79	*85.25*	*80.22*	81.65	72.27	84.96	79.30	84.12	77.45	80.72	69.74	58.21	52.38	59.22	59.22	48.91	78.90
Int	**87.94**	**88.21**	86.60	85.60	86.19	*85.07*	84.05	80.33	*86.39*	84.57	85.51	83.15	83.40	78.84	73.75	73.37	15.98	67.98	63.11	27.61
RAdr	**71.82**	**85.04**	69.63	82.83	70.58	84.29	59.20	73.56	69.10	82.73	62.84	75.85	63.14	77.95	0.00	0.00	128.59	3.29	4.78	155.82
LAdr	**72.68**	**84.15**	69.93	81.90	70.81	82.96	55.99	69.29	69.63	80.76	64.87	76.48	64.66	76.74	0.00	0.00	132.64	5.61	4.94	167.48
Rec	**83.10**	**79.59**	*81.53*	75.98	81.56	12.24	71.93	12.12	80.96	*74.85*	76.92	10.35	76.05	11.04	71.86	66.46	9.73	69.66	58.50	12.36
Bla	**94.86**	**90.88**	94.21	89.33	*94.08*	*89.01*	91.49	82.71	93.73	88.18	92.70	85.00	92.12	82.03	71.13	54.41	10.91	69.16	48.06	12.47
LFem	84.99	81.24	**92.27**	89.47	92.18	**89.61**	89.95	83.36	*91.79*	*88.34*	91.32	87.59	91.00	85.55	89.39	87.22	4.04	88.18	82.21	9.36
RFem	86.79	81.69	91.68	**88.50**	*91.66*	*88.30*	89.18	81.96	**91.74**	88.31	91.32	86.93	91.10	86.40	82.84	75.33	7.01	66.81	59.47	39.36
Pro	**93.01**	**88.96**	60.22	41.11	81.33	28.78	*84.75*	17.81	68.73	*62.06*	86.65	16.33	71.57	33.84	67.02	62.89	6.40	68.29	54.75	9.56
SVes	**88.23**	**88.50**	62.17	37.62	67.03	35.91	*77.48*	23.38	63.63	*37.04*	79.98	22.25	55.18	36.99	17.68	20.84	20.27	30.83	30.83	83.10
Mean	**86.65**	**86.87**	82.83	82.12	84.30	83.91	81.06	78.22	83.17	82.40	*83.23*	81.38	79.95	76.96	57.85	54.25	22.18	60.16	53.95	42.39

However, a direct numerical comparison alone does not fully capture the fundamental difference in methodology. The seven SOTA methods operate under a **class-index paradigm**: during both training and inference, each organ is identified by a fixed integer label. This means the target organ must be predefined and hardcoded into the model architecture. In contrast, our framework operates under a **text-guided retrieval paradigm**: the user specifies the target anatomy through a natural-language prompt, and the model retrieves and segments the corresponding region. This design offers several practical advantages:

- **Flexibility**: The model can respond to arbitrary textual descriptions without retraining. A new organ or anatomical variant can be queried simply by changing the input prompt.
- **No manual class mapping**: There is no need to maintain a fixed organ-to-index mapping, which simplifies deployment and reduces the risk of label mismatch errors.
- **Extensibility**: The text-guided paradigm naturally extends to organs or structures not seen during training, provided the language encoder captures sufficient semantic knowledge.

5.3.4.2 Per-Category Observations

Large organs. For anatomically prominent structures such as the liver, spleen, and kidneys, our model achieves DSC values in the range of 76–93%, which, while lower than the SOTA range of 95–97%, demonstrates that the text-guided retrieval cue is sufficient to activate and localize large anatomical targets. The left femur head (Stage 1 DSC = 89.39%) approaches SOTA-level performance, indicating that when the target has a distinctive shape and high contrast, the semantic query can achieve competitive localization.

Medium organs. Organs such as the pancreas (Stage 1 DSC = 65.27%), colon (58.21%), and intestine (73.75%) show moderate performance. These structures have irregular shapes and less distinct boundaries, which are challenging even for SOTA methods. The gap is wider here because class-index methods benefit from organ-specific learned features, whereas our text-guided approach must generalize across all organs using shared multimodal representations.

Small organs. The most challenging category reveals the sharpest contrast. Both adrenal glands achieve near-zero DSC in Stage 1 (0.00%), while the esophagus reaches only 1.33%. Stage 2 improves these substantially (esophagus: 39.26%, gallbladder: 40.25%), demonstrating that targeted optimization can enhance text-guided sensitivity for small targets. Nevertheless, SOTA methods achieve 70–83% DSC on these same organs, highlighting the inherent difficulty of localizing small structures through semantic queries alone.

Gender-specific organs. The prostate and seminal vesicles present an interesting case. Our Stage 1 achieves DSC of 67.02% and 17.68%, respectively, which are notably lower than the SOTA range. This gap likely reflects the limited representation of these organs in the training data and the inherent difficulty of text-guided retrieval for structures with highly variable appearance across patients.

5.3.5 Qualitative Results and Failure Cases

We will present some representative, Quite Successful Cases, Medium-Complexity Cases and Failure Cases as follows to qualitatively evaluate the performance of the proposed model:

5.3.5.1 Successful Cases

1. The Liver

   Prompt: Identify the Liver using spatial features.

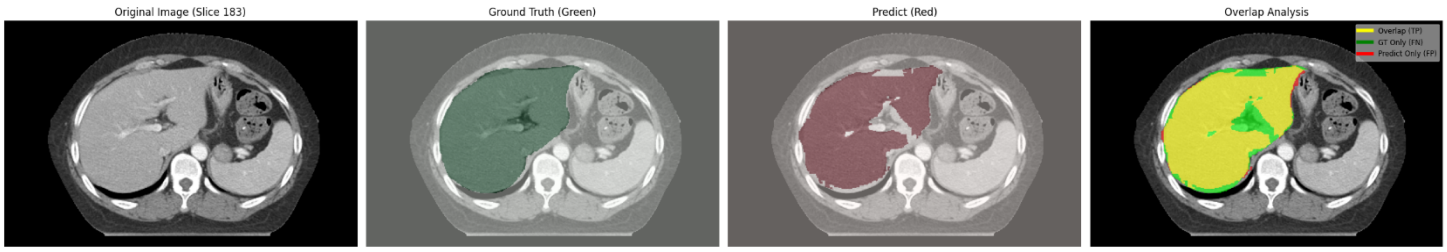


Figure 5.2: Liver segmentation results in Stage 1

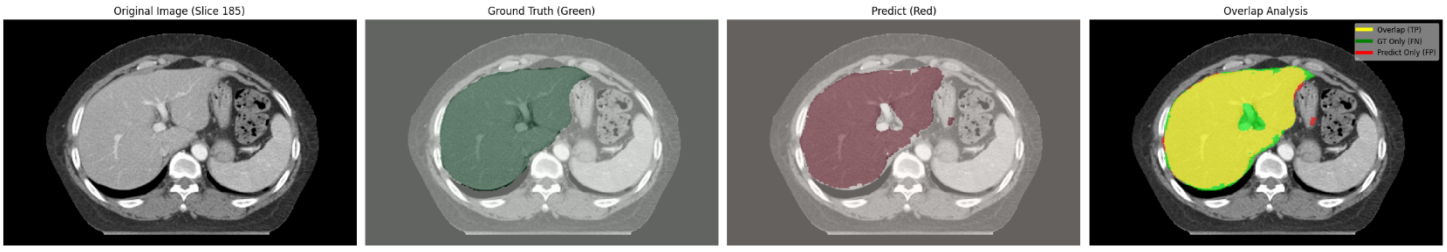


Figure 5.3: Liver segmentation results in Stage 2

The model produces a segmentation mask that closely follows the ground truth liver region. The predicted boundary is relatively smooth and anatomically plausible, with overlap over most of the organ volume. Minor boundary deviations may still appear, but overall the liver is one of the most stable and accurate cases, indicating that the framework can capture large, high-contrast anatomical structures effectively.

2. The Spleen

Prompt: Please segment the Spleen.

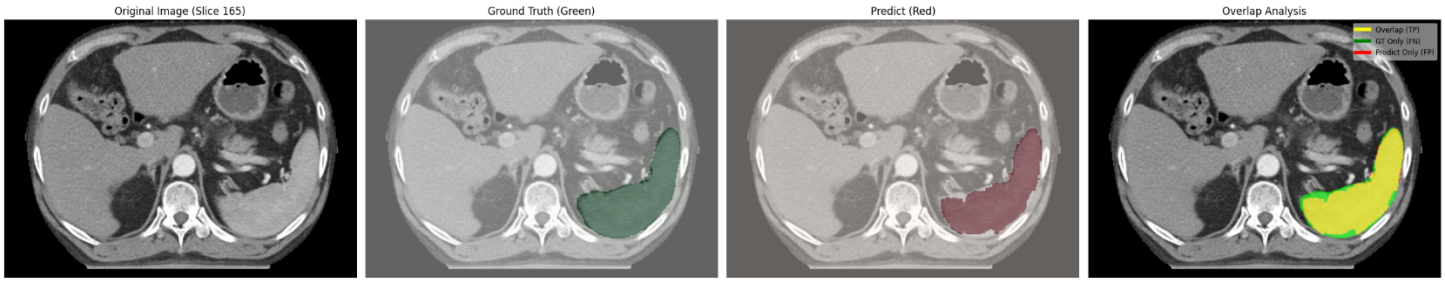


Figure 5.4: Spleen segmentation results in Stage 1

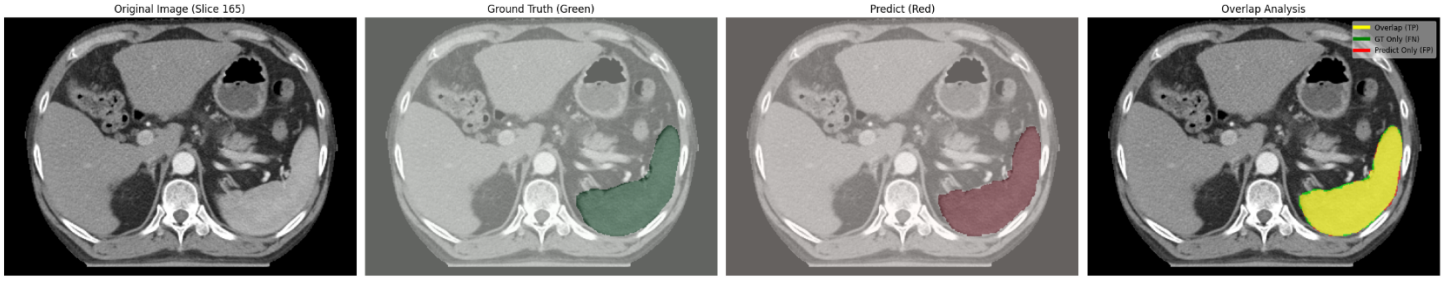


Figure 5.5: Spleen segmentation results in Stage 2

The predicted spleen mask is well aligned with the ground truth and generally preserves the correct organ shape. The boundary is mostly smooth, and the overlap is strong, showing that the model can localize this organ reliably. The prediction remains compact and coherent, suggesting that the spleen provides sufficiently strong visual signal for the current model to learn from.

5.3.5.2 Medium-Complexity Cases

1. The Pancreas

   Prompt: Limit the output to the Pancreas.

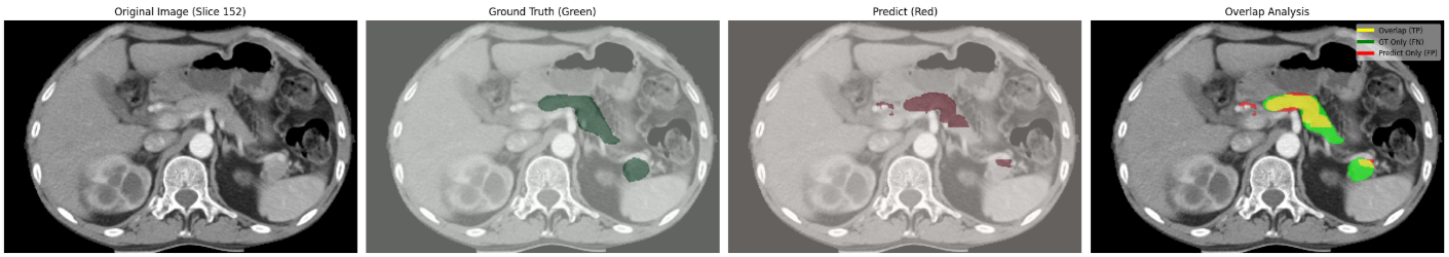


Figure 5.6: Pancreas segmentation results in Stage 1

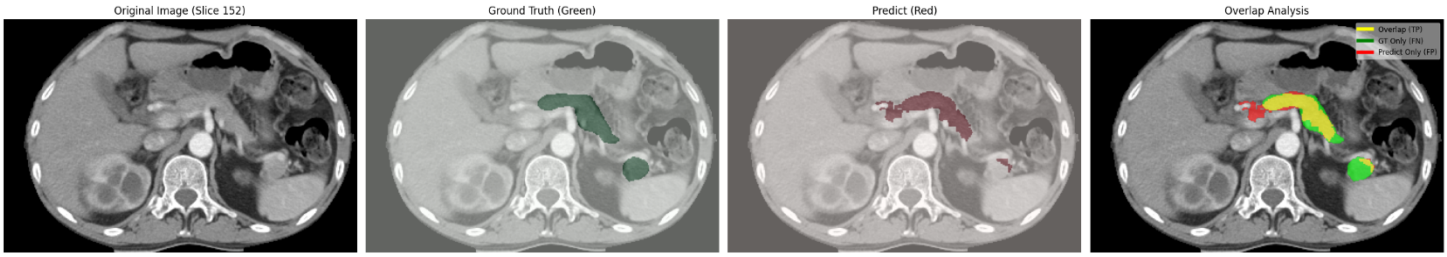


Figure 5.7: Pancreas segmentation results in Stage 2

The pancreas prediction captures the approximate anatomical location, but the segmentation is less stable than that of large organs. The boundary is not fully smooth, and the overlap with the ground truth is only moderate. This is because the pancreas has an irregular shape, and is more difficult to delineate accurately in 3D CT volumes.

2. The Colon

Prompt: The Colon.

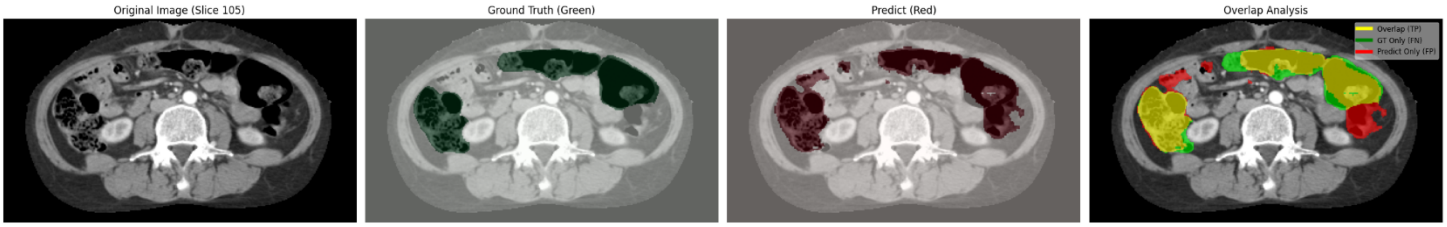


Figure 5.8: Colon segmentation results in Stage 1

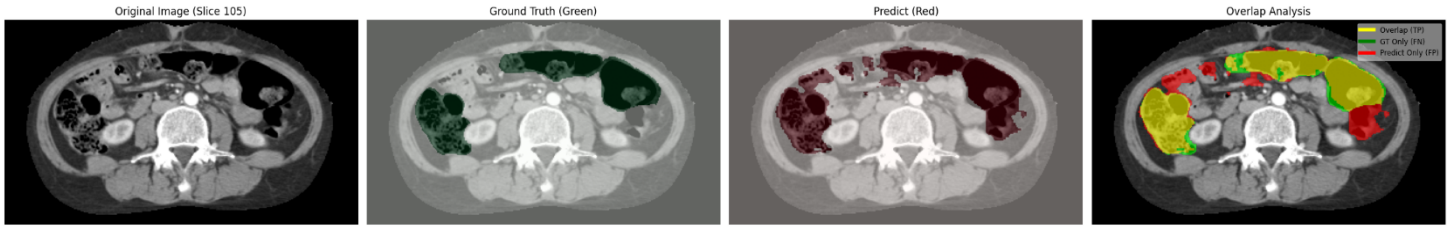


Figure 5.9: Colon segmentation results in Stage 2

The predicted colon region is generally located in the correct area, but the mask often shows incomplete coverage or boundary leakage. The overlap is acceptable, yet the contour is not fully clean due to the elongated and irregular morphology of the colon. This organ is more challenging than large compact organs because its appearance is less uniform and its boundary can be difficult to separate from adjacent structures.

5.3.5.3 Failure Cases

1. The Adrenal

   Prompt: The goal is to identify the Right adrenal gland.

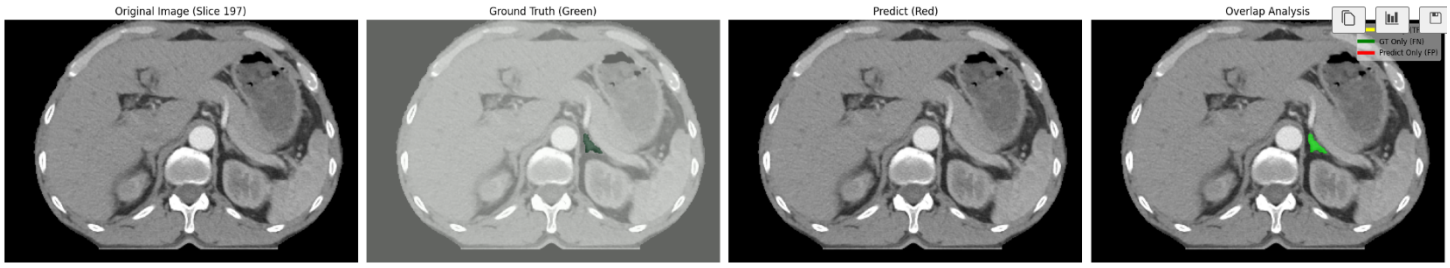


Figure 5.10: Right adrenal segmentation results in Stage 1

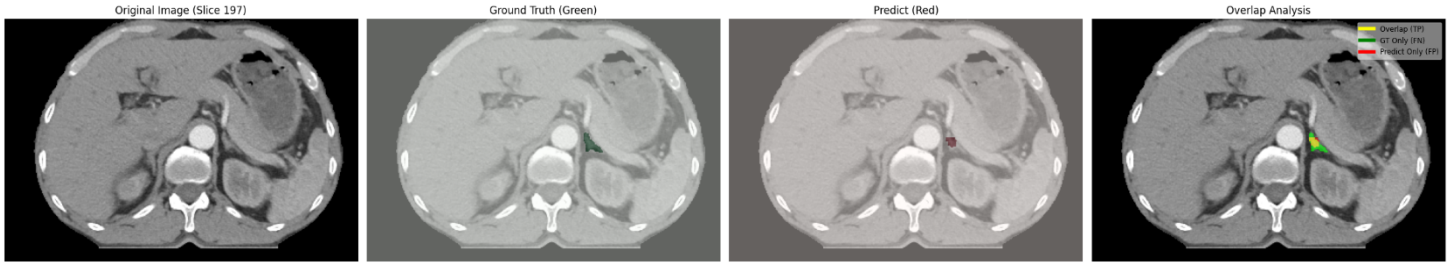


Figure 5.11: Right adrenal segmentation results in Stage 2

The adrenal is one of the most difficult structures for the model. In many cases, the predicted mask is empty or contains only a few sparse voxels, resulting in very poor overlap with the ground truth. This failure is mainly caused by the extremely small size of the adrenal gland, severe foreground/background imbalance, and the weak signal available within a full 3D patch.

2. The Gallbladder

Prompt: Identify the boundaries of the Gallbladder.

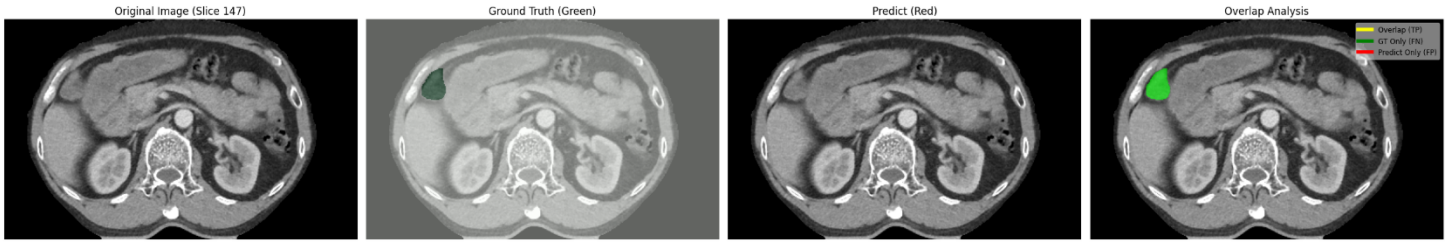


Figure 5.12: Gallbladder segmentation results in Stage 1

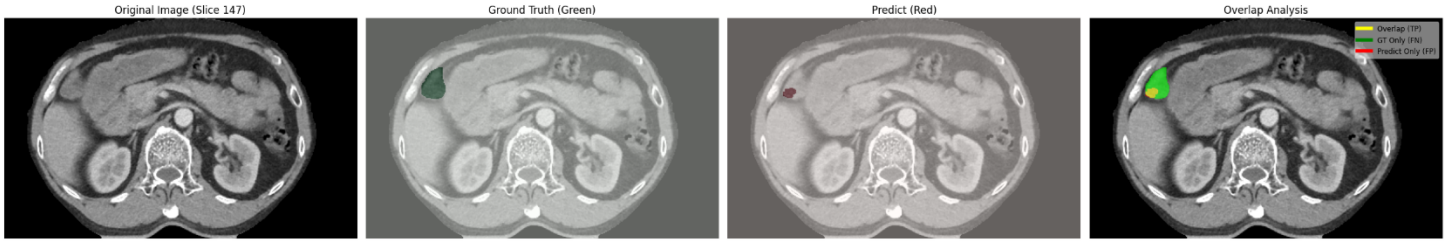


Figure 5.13: Gallbladder segmentation results in Stage 2

The gallbladder prediction is only partially correct and often remains incomplete or fragmented. This organ is difficult because it is relatively small, has low contrast with surrounding tissues, and does not provide a strong foreground signal during training, which makes reliable localization challenging.

5.3.5.4 Fragmentation Analysis

1. The Prostate

   Prompt: Select the Prostate.

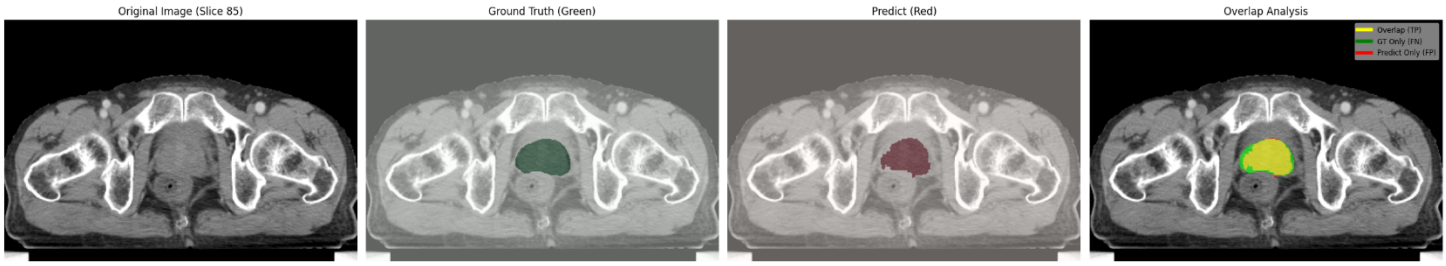


Figure 5.14: Qualitative prostate segmentation result

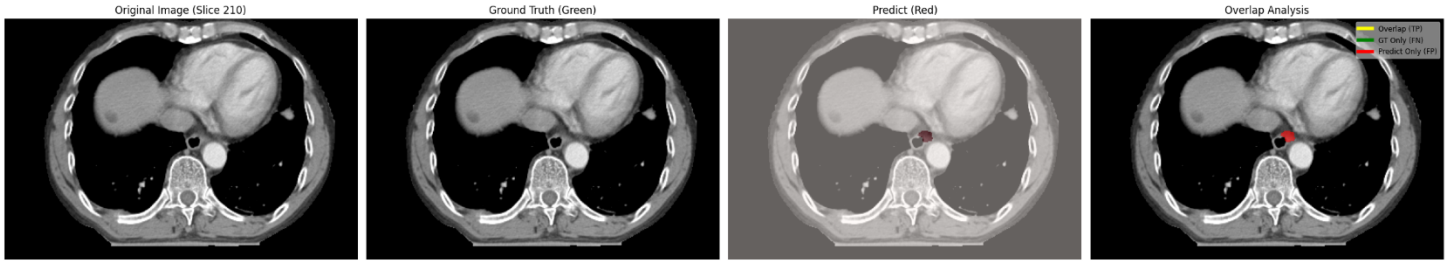


Figure 5.15: Fragmented prediction in prostate segmentation

2. The Bladder

Prompt: Locate the Bladder in the CT image.

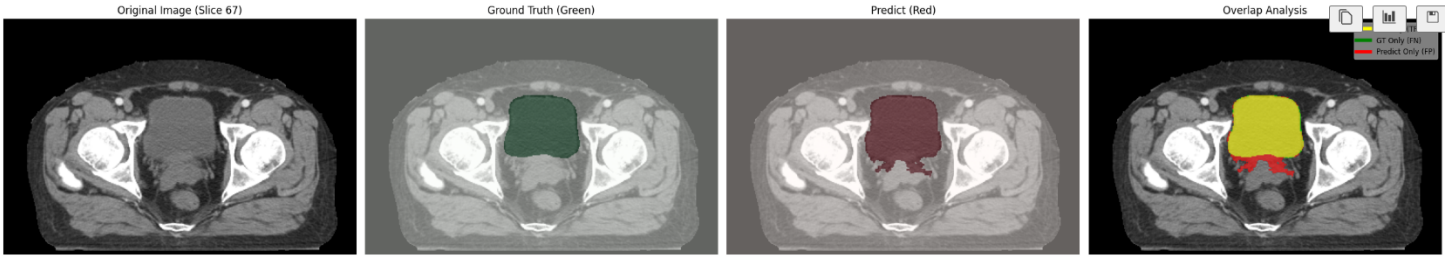


Figure 5.16: Qualitative bladder segmentation result

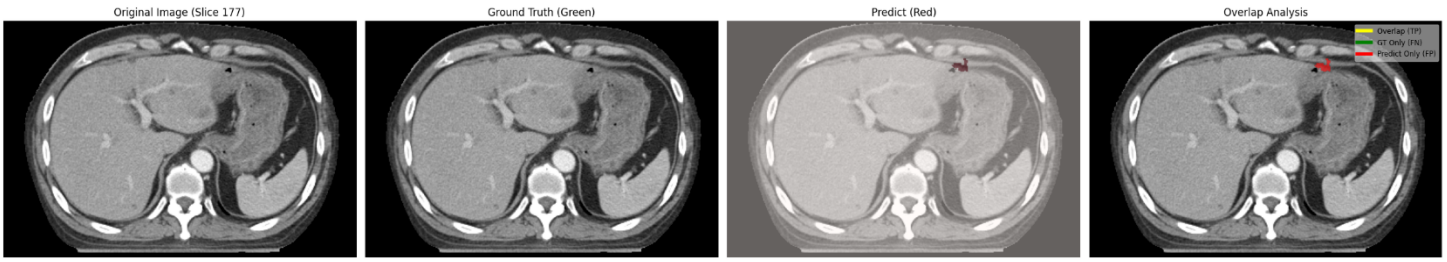


Figure 5.17: Fragmented prediction in bladder segmentation

In several cases, the model produces a reasonable main segmentation region but also generates small disconnected components far away from the target organ (about hundred slices far away). In such cases, the Dice score can still remain fairly acceptable because the main predicted region overlaps well with the ground truth. However, HD95 increases sharply because it is very sensitive to outlier voxels and distant false positives. This indicates that the model is not only affected by overlap quality, but also by prediction fragmentation and boundary instability. A simple connected-component post-processing step can suppress many of these small noisy blobs and improve the spatial consistency of the final mask.

5.3.6 Discussion

5.3.6.1 Strengths

One major strength of the proposed framework is its ability to use caption-like textual descriptions as effective semantic queries for anatomical localization. The quantitative results indicate that the model can consistently localize large organs such as the liver (DSC = 93.30%), spleen (89.10%), and left femur head (89.39%) with strong overlap accuracy. This suggests that the multimodal interaction between CT image features and text representations is sufficiently informative to guide segmentation when the target anatomy has clear structural characteristics and adequate spatial extent.

Another strength lies in the spatial stability achieved during the first optimization stage. Stage 1 produced relatively compact and coherent predictions, as reflected by low HD95 values (e.g., liver: 6.05, left femur head: 4.04, prostate: 6.40). This behavior shows that the framework is capable of conservative and reliable localization, which is an important property in text-guided medical segmentation because correct target activation alone is not sufficient if the predicted region is spatially unstable.

The experiments also demonstrate that the framework is responsive to targeted optimization strategies. After the second-stage refinement, several small and semantically difficult organs showed clear improvement: the esophagus improved from DSC of 1.33% to 39.26%, the gallbladder from 8.14% to 40.25%, with both organs simultaneously improving in HD95. This indicates that the proposed model is flexible and that its semantic sensitivity can be enhanced through appropriate changes in data emphasis, loss design, and preprocessing resolution without requiring a complete redesign of the architecture.

A key distinguishing feature of the proposed framework is the **text-guided retrieval paradigm**. Unlike conventional SOTA methods that require a fixed class-index mapping between organs and integer labels, our approach enables **prompt-based organ retrieval**: the user specifies the target anatomy through a natural-language description, and the model automatically retrieves and segments the corresponding region. This eliminates the need for manual organ selection or predefined label dictionaries, making the framework more flexible and adaptable to new anatomical targets. While current performance is lower than fully supervised methods (mean DSC of 60.16% vs. 86.65% for nnU-Net), this gap reflects the inherent challenge of multimodal alignment rather than a fundamental limitation of the approach. The framework demonstrates that semantic queries can drive meaningful anatomical localization, opening a pathway toward interactive, language-driven medical image analysis.

5.3.6.2 Weaknesses

The main weakness of the proposed framework is its limited robustness on very small anatomical structures. In Stage 1, several small organs were missed entirely (adrenal glands: 0.00% DSC, esophagus: 1.33%), indicating that semantic conditioning alone was not sufficient to overcome severe voxel imbalance. While Stage 2 improved these cases, it introduced a trade-off: mean HD95 increased from 22.18 to 42.39, and some predictions contained disconnected fragments far from the true anatomy. This fragmentation reduces practical reliability despite acceptable DSC values.

Additionally, as shown in the SOTA comparison (Table 5.11), a substantial performance gap remains between our text-guided approach (mean DSC = 60.16%) and fully supervised methods (nnU-Net mean DSC = 86.65%). This gap is inherent to the different task formulations but highlights the need for further advances in multimodal alignment and spatial regularization.

5.3.6.3 Analysis

Taken together, the results suggest that the proposed framework already captures a meaningful semantic correspondence between text and anatomy, but it does not yet balance overlap improvement and spatial reliability equally well. In other words, the model is able to respond to caption-like anatomical queries, yet the quality of that response depends strongly on the optimization regime and the anatomical scale of the target structure.

The central finding is therefore a clear trade-off between DSC improvement for difficult small organs and HD95 changes across the organ set. Stage 1 favors conservative and stable localization: it performs strongly on large organs and maintains relatively low HD95 values, but it remains insufficient for several very small and semantically difficult targets. In contrast, Stage 2 improves DSC for several hard small organs by making the model more sensitive to underrepresented foreground regions. Notably, this sensitivity gain produces mixed effects on HD95: some organs (esophagus, gallbladder) improve in both DSC and HD95, while others (spleen, seminal vesicles) experience significant HD95 degradation.

When positioned against SOTA methods, the results demonstrate that **text-guided organ retrieval is a viable but challenging paradigm**. The proposed framework successfully demonstrates that a natural-language prompt can drive anatomical segmentation without manual class mapping or organ-index selection. While the numerical performance is currently below fully supervised methods, the flexibility, extensibility, and user-friendliness of the prompt-based approach represent a meaningful contribution toward interactive medical image analysis.

This trade-off is important because it shows that better overlap does not au-

tomatically lead to better anatomical segmentation quality. A model may recover more of the intended target and thus obtain higher DSC, while still producing anatomically implausible or spatially scattered predictions that worsen HD95. Future work should therefore aim to preserve the DSC gains achieved in Stage 2 while reducing HD95 inflation and fragmentation, for example through stronger structural regularization, better hard-sample control, or more selective post-processing for false-positive suppression. Additionally, bridging the gap with SOTA methods could involve incorporating organ-specific attention mechanisms guided by the text embedding, or employing multi-scale training strategies that better preserve small-organ features.

5.3.7 Summary

This chapter evaluated the proposed model, a combination of SwinUNETR and BioBERT, through two training stages and compared it against seven state-of-the-art methods on the RAOS benchmark.

Stage 1 achieved stable performance on large organs such as the liver (DSC = 93.30%), spleen (89.10%), and left femur head (89.39%), with low HD95 values indicating compact and reliable segmentation results. Stage 2 improved the recovery of several difficult small organs, including the esophagus (from 1.33% to 39.26% DSC), gallbladder (from 8.14% to 40.25%), and seminal vesicles (from 17.68% to 30.83%), demonstrating that targeted optimization can increase semantic sensitivity. However, this improvement came with a trade-off: some predictions showed increased fragmentation, and the mean HD95 increased from 22.18 to 42.39. Notably, certain small organs such as the esophagus and gallbladder improved in both DSC and HD95 simultaneously.

The comparison with SOTA methods reveals a substantial performance gap: our best configuration (Stage 2) achieves a mean DSC of 60.16% compared with 86.65% for the top-performing nnU-Net. However, this comparison must be interpreted in the context of fundamentally different task formulations. While SOTA methods rely on fixed class-index mappings and organ-specific learned features, our framework performs text-guided anatomical retrieval, enabling flexible, prompt-based organ selection without manual class assignment. This paradigm offers practical advantages in extensibility and user interaction, representing a step toward interactive, language-driven medical image analysis.

Overall, the results suggest that the proposed framework is capable of effective caption-like anatomical localization, but its performance still depends strongly on organ size and optimization strategy. Future improvements should focus on preserving the DSC gains for small organs while reducing fragmentation, improving boundary compactness, and incorporating mechanisms to narrow the gap with fully supervised methods.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project investigated how generated caption-like anatomical descriptions can support multimodal semantic localization in three-dimensional CT volumes. The central research objective was to determine whether generated anatomical captions could serve as semantic retrieval cues that guide image–text alignment and support anatomical region recovery within volumetric medical images. Within the conceptual framing of this thesis, text-to-image retrieval can therefore be interpreted not as external image ranking, but as caption-conditioned anatomical target recovery inside 3D medical data, where the resulting voxel mask serves as a retrieval-oriented localization output.

To study this problem, the thesis developed a multimodal framework that combines a SwinUNETR image encoder, a BioBERT text encoder, multi-head cross-attention fusion, and a 3D U-Net-style decoder. Generated single-organ and multi-organ anatomical prompts, together with clean and noisy caption variants, were used as semantic query inputs. Under this formulation, the generated anatomical captions can be interpreted as structured semantic queries that help the model associate linguistic intent with anatomically relevant regions in CT volumes. The results suggest that this image–text semantic alignment supports multimodal anatomical retrieval and provides evidence that caption-based semantic querying can recover organ-specific regions in a coherent voxel-level form.

The experimental results in Chapter 6 further support this interpretation. In Stage 1, evaluated at epoch 126 with `DiceCELoss` and a preprocessing pixdim of $1.5 \times 1.5 \times 1.5$, the framework achieved an overall Dice Score of 0.6064 and an HD95 of 31.9095. These results indicate stable and spatially compact localization

for anatomically prominent structures such as the liver, spleen, kidneys, stomach, and femur heads. Stage 1 therefore suggests that caption-conditioned anatomical retrieval was already feasible for large organs with clearer semantic and spatial signatures.

In Stage 2, optimization was extended to epoch 176 with approximately two-fold greater exposure to small-organ samples, a finer preprocessing pixdim of $1.2 \times 1.2 \times 1.2$, and `DiceFocalLoss`. The overall Dice Score increased slightly to 0.6092, while several difficult small organs showed substantial improvements: the esophagus rose from 0.0151 to 0.4532, the seminal vesicles from 0.1473 to 0.4601, the gallbladder from 0.2161 to 0.3735, the left adrenal gland from 0.0000 to 0.0561, and the right adrenal gland from 0.0000 to 0.0906. These findings indicate that stronger semantic target activation can improve caption-conditioned anatomical recovery for underrepresented and spatially challenging structures.

At the same time, the Stage 2 results suggest that optimization strategies that improve small-organ semantic sensitivity may also reduce spatial compactness. The increase in overall HD95 from 31.9095 to 136.0575 indicates that stronger foreground activation alone does not guarantee stable localization boundaries, particularly when distant false-positive fragments remain after inference. This should be interpreted as an important secondary finding rather than the central conclusion of the thesis.

Overall, this thesis demonstrates the feasibility of using generated caption-like anatomical descriptions as semantic retrieval cues for multimodal anatomical localization in 3D medical images. Although the present study does not constitute a formal large-scale retrieval benchmark or a full caption-generation evaluation, the findings suggest that generated anatomical prompts can support image–text semantic alignment, caption-conditioned anatomical retrieval, and retrieval-oriented localization behavior in a manner consistent with the thesis title. Further validation is required, but the framework provides a useful basis for future research on multimodal medical image analysis driven by semantic querying.

6.2 Summary of Contributions

This thesis made a focused contribution to multimodal medical image analysis by investigating how generated anatomical captions can guide semantic querying and anatomical region recovery in 3D CT volumes. The study did not aim to claim a full retrieval benchmark; rather, it developed and evaluated a caption-conditioned framework whose outputs can be interpreted as retrieval-oriented localization results. The main contributions are summarized as follows:

- Developed a multimodal semantic localization framework that integrates a SwinUNETR image encoder, a BioBERT text encoder, multi-head cross-

attention fusion, and a 3D U-Net-style decoder for caption-conditioned anatomical retrieval in volumetric CT data.

- Generated diverse caption-like anatomical prompts for single-organ and multi-organ semantic querying, including short commands, localization-oriented prompts, mask-generation instructions, exclusion instructions, and longer anatomical descriptions.
- Included noisy caption variants with character-level corruption to introduce robustness-oriented semantic querying and to examine how imperfect textual input may influence anatomical region recovery.
- Implemented cross-modal attention to align volumetric image features with caption-derived semantic representations, thereby supporting semantic target activation and anatomically relevant region recovery.
- Evaluated stage-wise optimization behavior on the RAOS dataset across 19 anatomical organs using Dice Score and HD95 as complementary measures of overlap quality and spatial compactness.
- Analyzed how optimization changes affected semantic target activation, showing that Stage 2 improved several difficult small organs while Stage 1 remained stronger for stable and compact localization of large anatomical structures.
- Identified a trade-off between improved small-organ retrieval sensitivity and reduced spatial compactness, especially when fragmentation increased HD95 despite modest gains in overlap.

6.3 Limitations

Although the results are encouraging, the current framework still encounters several limitations that should be considered when interpreting the findings. First, the present evaluation remains limited by the scope of comparison. While the study examined stage-wise optimization behavior in detail, it does not yet include a broad set of external baselines such as 3D U-Net, SwinUNETR without caption input, or stronger medical vision–language models. Similarly, the present study does not yet include a large-scale formal retrieval ranking benchmark, so the retrieval interpretation of the model is grounded primarily in multimodal localization behavior rather than in ranked image retrieval experiments.

Second, the empirical validation remains limited in statistical depth. The reported results provide useful evidence regarding Dice and HD95 trends, but large-scale statistical testing across multiple random seeds, broader robustness protocols,

and expert clinical review were outside the scope of this thesis. Further validation is therefore required before stronger general claims can be made about generalization or clinical usefulness.

Third, small organs remain challenging despite the gains observed in Stage 2. The esophagus, gallbladder, seminal vesicles, and both adrenal glands benefited from stronger semantic target activation, but the adrenal organs in particular still achieved relatively low Dice values after optimization. This indicates that caption-conditioned anatomical retrieval for very small structures remains sensitive to voxel imbalance, anatomical variability, and fine-scale boundary ambiguity.

Fourth, the Stage 2 optimization strategy introduced a clear compactness limitation. Although overlap improved slightly and small-organ recovery became more responsive to semantic queries, HD95 increased substantially because fragmented predictions and distant false-positive components remained after inference. This suggests that the current framework can recover the intended anatomical target more sensitively under some settings, but still requires stronger control of boundary stability and spatial coherence.

Fifth, the thesis included noisy captions and multi-organ captions to broaden the semantic-query setting, but deeper robustness analysis is still needed. The present study does not yet provide a fully systematic comparison across clean captions, noisy captions, paraphrased prompts, ambiguous descriptions, and more complex multi-organ query types. As a result, the robustness of caption-based semantic querying remains only partially characterized.

Finally, the framework is computationally demanding because it operates on 3D medical volumes with multimodal feature extraction and fusion. Training and inference therefore require substantial memory and processing time. In addition, the experimental evidence is currently limited to the RAOS dataset, so broader cross-dataset generalization remains an open question for future work.

6.4 Future Research Directions

The findings of this thesis suggest several promising directions for extending caption-conditioned anatomical retrieval in 3D medical images. Future work should strengthen the empirical basis of the framework, improve the robustness of semantic querying, and better balance semantic target activation with spatial compactness. In particular, further experiments are required to determine how generated anatomical captions, multimodal alignment strategies, and optimization choices interact across broader datasets, model families, and clinical scenarios.

6.4.1 Expanding Model Comparisons and Retrieval Baselines

A natural next step is to broaden the comparison setting so that the role of generated caption-like prompts can be evaluated more precisely. Future studies should compare the current framework with stronger and more diverse baselines, including 3D U-Net, SwinUNETR without caption input, and simpler fusion strategies in which organ labels or shallow caption embeddings are injected without full cross-modal attention. Such comparisons would help separate the contributions of the image encoder, the text encoder, and the fusion module, and would clarify whether generated anatomical captions provide measurable benefits over simpler organ-name supervision alone.

It would also be valuable to compare the current framework with stronger medical vision–language models and, when feasible, with retrieval-style baselines that explicitly evaluate semantic matching between captions and medical volumes. Although the present study interprets the voxel mask as a retrieval-oriented localization output, future work could define a more formal caption-conditioned retrieval benchmark in which localization quality is analyzed together with query sensitivity, ranking behavior, or region-level matching criteria. This would strengthen the connection between multimodal anatomical retrieval and the broader literature on text-to-image retrieval.

Further comparison should also examine the linguistic structure of the generated prompts themselves. Single-organ captions, multi-organ captions, noisy captions, short commands, longer anatomical descriptions, and simple labels should be evaluated under a common protocol to determine which forms of caption-based semantic querying are most informative. Such experiments would provide clearer evidence regarding whether richer generated prompts improve semantic target activation, whether noisy prompts reduce localization reliability, and whether multi-organ captions encourage more complex retrieval-oriented behavior inside a 3D volume.

Beyond architectural comparisons, future work should expand the evaluation setting to additional datasets and clinical text sources. Cross-dataset experiments would help determine how robust the learned image–text alignment remains across institutions and imaging distributions. Real clinical reporting language could also be incorporated as a more realistic source of caption-like queries, while expert review of retrieved anatomical regions would provide stronger validation of the medical plausibility of the localization outputs.

6.4.2 Fine-tuning and Optimization of Multimodal Components

A second major direction is to improve the optimization of the multimodal components themselves. In the current study, frozen encoders provided stable training and a clear experimental setup, but future work should investigate fine-tuning BioBERT for anatomical caption understanding and selectively unfreezing parts of SwinUNETR to improve image–text adaptation. Systematic comparisons between frozen and fine-tuned encoders would help determine whether stronger task-specific alignment can improve semantic region recovery without sacrificing stability. The cross-attention fusion module could also be trained more deeply or redesigned to support richer interactions between caption tokens and spatial image tokens.

Loss design is another important opportunity for improvement. The comparison between `DiceCELoss` and `DiceFocalLoss` suggests that optimization strongly influences the balance between semantic sensitivity and spatial compactness. Future work should therefore examine `TverskyLoss`, weighted Dice loss, adaptive organ-size weighting, and hybrid objectives that combine overlap, focal emphasis, and boundary regularization. Boundary-aware loss functions, fragmentation-aware penalties, and topology-preserving constraints may be especially useful for reducing disconnected false-positive components while preserving the gains observed for small semantic targets.

Because the present results show that small-organ recovery improved while HD95 degraded, future optimization should explicitly aim to balance Dice improvement with HD95 stability. Organ-aware weighting schemes could place additional emphasis on underrepresented anatomical structures without allowing sensitivity to spread excessively beyond the intended target. Likewise, adaptive weighting based on organ size, boundary complexity, or retrieval uncertainty may help stabilize caption-conditioned anatomical retrieval across diverse structures.

Post-processing and spatial regularization should also be explored more systematically. Connected component filtering, largest-component selection, anatomical priors, shape constraints, and learned spatial refinement modules may help reduce fragmentation and improve compactness. Such strategies could make retrieval-oriented localization outputs more coherent, especially for small organs and multi-organ prompts in which semantic activation may be spatially diffuse.

Overall, future research should aim not only to improve overlap scores, but also to refine how multimodal semantic alignment translates into stable anatomical region recovery. A stronger balance among caption understanding, semantic target activation, compact localization, and robustness to noisy or complex prompts would substantially strengthen the role of generated anatomical captions in 3D medical image analysis.

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